Multiscale 3-D–2-D Mixed CNN and Lightweight Attention-Free Transformer for Hyperspectral and LiDAR Classification

Le Sun, *Member, IEEE*, Xinyu Wang, Yuhui Zheng, *Senior Member, IEEE*, Zebin Wu, *Senior Member, IEEE*, and Liyong Fu

***Abstract*— The effective combination of hyperspectral image (HSI) and light detection and ranging (LiDAR) data can be used for land cover classification. Recently, deep-learning-based classification methods, especially those using transformer networks, have achieved remarkable success. However, deep learning classification methods for multisource data still encounter various technical challenges, such as comprehensive utilization of multiscale information, lightweight network design, and efficient fusion strategies for heterogeneous data. To address these challenges, we propose a novel and efficient deep neural network, namely, multiscale 3-D–2-D mixed CNN feature extraction and multisource data lightweight attention-free fusion network (M2FNet) based on CNN and transformer. Through end-to-end training, this network effectively combines heterogeneous information from multiple sources, leading to improved performance in joint classification. Specifically, M2FNet uses a multiscale 3-D–2-D mixed CNN design to extract both the spatial–spectral features of HSI and the depth-based elevation features of LiDAR data. Subsequently, the extracted features are fed into a novel encoder comprising a feature enhancement (FE) module, designed with mathematical morphology and a dilated convolutional module derived from the self-attention of the conventional transformer encoder (DConvformer), which plays a crucial role in integrating multisource information within the network. The well-designed architecture enables the network to acquire multiscale depth and high-order features, significantly reducing the number of training parameters. Comparative experimental results and ablation studies demonstrate that M2FNet outperforms other advanced methods. The source code is publicly available at https://github.com/cupid6868/M2FNet.git.**

***Index Terms*— Convolutional neural network (CNN), hyperspectral image (HSI), joint classification, light detection and ranging (LiDAR) data, multiscale, transformer.**

I. Introduction

LAND cover classification of remote sensing images [1], [2], [3], [4] has been widely applied in many fields, such as mineral exploration [5], environmental monitoring [6], urban development and planning [7], forest management [8], precision agriculture [9], food security [10], medical diagnosis [11], and target detection [12], [13]. Hyperspectral image (HSI) can capture the reflectance data of the surface in multiple continuous spectral bands [14] and is often used as the main data source for classification tasks due to its rich spectral and spatial information. In HSIs, the spectral signals from different categories exhibit similarity or overlap, which presents a challenge in accurately distinguishing or identifying unique features. This confusion can be attributed to the similarity in spectral characteristics of materials, as seen in the difficulty in classifying structures such as tall buildings and roads, both made of concrete. Although the classification technology has achieved satisfactory results, how to solve the spectral-like confusion problem [15], [16] is still a core problem that needs to be considered urgently. In recent years, light detection and ranging (LiDAR) data can accurately determine the height of two objects with similar spectral curves and thus can provide valuable information to directly distinguish between them [17]. Due to its valuable spatial information and elevation data, the joint classification of HSI and LiDAR data has attracted the interest of researchers [18], [19].

In the early stages of classifying remote sensing data, the predominant approach was to rely exclusively on a single type of data. For example, in HSI classification, the focus was primarily on using HSI data alone. When it comes to the task of HSI classification, we cannot ignore those classical machine learning algorithms that play a key role in solving this problem. These traditional methods, such as support vector machine (SVM) [20], [21], random forest (RF) [22], k-nearest neighbors (k-NN) [23], principal component analysis (PCA) [24], linear discriminant analysis (LDA) [25], and extreme learning machine (ELM) [26], have brought a rich selection of techniques to the field of HSI classification while retaining their practicality. However, the limitations of the traditional

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Le Sun is with the School of Computer Science, Jiangsu Collaborative Innovation Center of Atmospheric Environment and Equipment Technology (CICAEET), Nanjing University of Information Science and Technology, Nanjing 210044, China (e-mail: sunlecncom@163.com).

Xinyu Wang is with the School of Computer Science, Nanjing University of Information Science and Technology, Nanjing 210044, China (e-mail: wxysun88@126.com).

Yuhui Zheng is with the College of Computer, Qinghai Normal University, Xining 810008, China, and also with the School of Computer Science, Nanjing University of Information Science and Technology, Nanjing 210044, China (e-mail: zhengyh@vip.126.com).

Zebin Wu is with the School of Computer Engineering, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: zebin.wu@gmail.com).

Liyong Fu is with the Institute of Forest Resource Information Techniques, Chinese Academy of Forestry, Beijing 100091, China (e-mail: fuliyong840909@163.com).

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machine learning algorithms using single HSIs have led to more and more research on the joint classification of HSI and LiDAR data. Colgan et al. [27] used the fusion of HSIs and LiDAR data to use a two-stage SVM classifier for predicting the species of each tree. Huang and Zhu [28] proposed an integration method based on RF, which combined the information of spectrum, elevation, and their corresponding texture to obtain a better final classification result. Jia et al. [29] proposed a classification method for HSIs and LiDAR data based on multifeature superpixel-level decision fusion, which designed and used kernel PCA (KPCA), 2-D and 3-D Gabor filters, and RF classifier.

The traditional machine learning methods require manual feature engineering [30]. They have limited ability to model nonlinear relationships, are sensitive to noise and outliers, require more parameter adjustments, and are limited when dealing with large-scale complex data. With the rapid development of deep learning, numerous methods have emerged in the field of HSI classification [31], [32], [33]. Initially, the significant impact of convolutional deep neural networks was primarily introduced to address classification problems, including methods such as 1-D CNN [34], 2-D CNN [35], and 3-D CNN [36]. However, the limitations of singular convolutions in extracting features have gradually become apparent. He et al. [37] proposed a multiscale 3-D deep convolutional neural network (M3D-CNN) for HSI classification, which simultaneously learned both 2-D multiscale spatial features and 1-D spectral features from HSI data in an end-to-end manner. The 3-D architecture proposed in [38] not only accurately classifies the HSIs but also establishes a deep comprehension of the images at a low cost. One valuable consequence was the efficient optimization of deep networks on small-sized annotated datasets, resulting in reduced data costs. However, the emergence of 3-D CNN in HSI classification also revealed its shortcomings. Challenges arise in dealing with large-scale and multiscale data, as well as capturing long-distance dependencies. Following the emergence of the attention mechanism in natural language processing [39] and image processing [40], the attention module has been widely applied to various models. Guo et al. [41] proposed the dual-view spectral and global spatial feature fusion network (DSGSF) to extract spatial–spectral features for HSI classification. The network consists of a spatial subnetwork with channel attention and spatial attention and a spectral subnetwork with viewpoint attention. Zhou et al. [42] introduced the attention multihop graph and multiscale convolutional fusion network (AMGCFN), which consists of two subnetworks: multiscale complete CNN and multihop graph convolution network (GCN). This model enhances classification accuracy. Chen et al. [43] developed a classification framework that includes a spectral–spatial diffusion module and an attention-based classification module. The diffusion features enhance cross-sample perception by reconstructing the distribution, leading to improved classification performance. More recently, a model known as the vision transformer (ViT) [44] has made a significant impact in the field of computer vision. This has led to a surge of research on transformers in the domain of HSI classification. In this regard, Hong et al. [45] introduced a novel backbone network called SpectralFormer. Unlike the traditional transformers that focus on bandwise representations, SpectralFormer can effectively capture spectrally local sequence information from neighboring bands of HSIs, resulting in groupwise spectral embeddings. Another approach, proposed by Sun et al. [46], was the spectral spatial feature tokenization transformer (SSFTT), which leveraged a combination of 3-D-CNNs and 2-D-CNNs to extract shallow spectral and spatial features. These features were then fed into a carefully designed transformer encoder after undergoing feature transformation using Gaussian weighting. In addition, Fang et al. [47] used a three-branch network architecture to maximize the utilization of ultrasmall training samples. They devised a series of multiattention modules that use lightweight transformers for encoding and classification. Furthermore, to prevent limiting the transformer's ability to learn multiscale features, Tu et al. [48] introduced a local semantic feature aggregation-based transformer (LSFAT) architecture. This framework successfully captured multiscale features and achieved excellent classification performance. In addition, studies have been carried out in various areas, including image transfer networks [49], semi-supervised learning [50], adversarial sample defense [51], and others. However, using single HSI data for classification has gradually revealed limitations, such as spectral confusion caused by shared building materials between roads and structures. Therefore, fusing HSI with LiDAR data in classification overcomes these constraints and greatly improves classification accuracy.

Deep-learning-based algorithms have shown great potential in the joint classification of HSI and LiDAR data [52], which is a topic of high concern. Feng et al. [53] proposed an efficient linear self-attention fusion (LSAF) model for the joint classification of HSI and LiDAR data. The proposed method included a feature extraction module, an attention module, and a fusion module. Hang et al. [54] proposed an efficient and effective framework, called CoupledCNN, to fuse HSI and LiDAR data using two coupled convolutional neural networks. During the fusion phase, feature-level and decision-level fusion methods were simultaneously used to effectively integrate these heterogeneous features. Zhao et al. [55] proposed a new hierarchical CNN and transformer network (HCT) and designed a novel cross-token attention fusion encoder to effectively fuse HSI and LiDAR data, thereby improving classification performance. Wang et al. [56] proposed a nearest-neighbor-based contrastive learning network (NNCNet). Specifically, they proposed a data augmentation scheme based on a nearest-neighbor approach to leverage enhanced semantic relationships among neighboring regions. In addition, they designed a bilinear attention module to capture the second-order and high-order feature interactions between the HSI and LiDAR data. Lu et al. [57] proposed an end-to-end deep learning model based on coupled adversarial learning classification (CALC). Specifically, they designed a coupled adversarial feature learning subnetwork and a multilevel feature fusion classification subnetwork for the purpose of HSI and LiDAR data fusion classification.

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Despite the significant progress made by deep learning methods in classifying HSI and LiDAR data, there are still challenges and limitations. First, the comprehensive exploitation of multiscale information requires the adept handling of data from diverse sensors with varying spatial resolutions [58], necessitating a more efficient feature extraction network that effectively captures both fine-grained local details and broader contextual information [59], [60]. Second, the lightweight design of neural networks is crucial, particularly in resource-constrained scenarios or real-time applications. Achieving high classification accuracy while minimizing computational complexity involves a delicate balance between model architecture, compression methods, and quantization [61], [62]. Finally, an efficient fusion strategy involving HSI and LiDAR data requires proficient data integration techniques to fully leverage the unique characteristics of each data source, thereby achieving precise and comprehensive scene classification. When working with multisource remote sensing data, addressing these challenges is of paramount importance to ensure both accuracy and efficiency in classification tasks.

To address the aforementioned challenges, this article introduces a comprehensive framework for the joint classification of HSI and LiDAR data. The proposed framework incorporates multiscale 3-D–2-D mixed CNN feature extraction, to capture spectral, spatial, and elevation features from HSI and LiDAR data. Within the framework, a DConvformer module is proposed, and the feature enhancement (FE) component in this module uses morphological erosion and dilation operations based on mathematical morphology [63], [64], to further strengthen the extracted features. This enhancement aims to strengthen the interaction between tokens, thereby enhancing the representation of structural and shape information. It is worth noting that in our proposed DConvformer module, we have replaced the traditional self-attention mechanism with the deep dilated convolution component. This substitution not only improves classification accuracy but also significantly reduces the number of model training parameters.

The main contributions of this article are summarized as below.

1) A dual-branch network has been designed, leveraging the distinctive capabilities of a multiscale 3-D–2-D mixed CNN. This network facilitates the effective joint classification of spectral–spatial–elevation features in HSI-LIDAR data, resulting in a significant improvement in fusion classification accuracy.
2) A novel DConvformer module is introduced for effectively exploring deep features and fusing heterogeneous information. It uses a morphological FE module and a deep dilated convolution module to efficiently extract morphological features and explore the deep spectral–spatial–elevation features of the data. The module enables efficient interaction and integration of features derived from the multiscale CNN. In addition, it leverages interactions between tokens to effectively use the complementary advantages provided by various data sources.
3) A series of lightweight operations have been used in the network. For instance, grouped convolution is used in the multiscale 3-D–2-D mixed CNN module. In addition, the self-attention module is replaced with the deep dilated convolution module, significantly reducing the computational complexity of the model. Extensive experiments on three datasets demonstrate the effectiveness of the proposed method.

The remainder of this article is organized as follows. Section II describes the proposed framework. In Section III, the experimental datasets, experimental settings, and analysis of the classification results are presented. The conclusions of this work and some discussions on possible future research directions are provided in Section IV.

II. Methodology

The proposed framework for the joint classification of HSI and LiDAR data is described in detail in this chapter. The five stages that make up the framework are HSI and LiDAR data preprocessing, HSI and LiDAR feature learning via multiscale 3-D–2-D mixed CNN, feature tokenization and position embedding, DConvformer encoder, and classifier head.

A. Overall Architecture

The purpose of the proposed multisource data lightweight attention-free fusion network (M2FNet) framework is to construct a hierarchical fusion network using a new framework and efficient transformer with CNN layers to achieve the fusion and classification of HSI and LiDAR data. The overall architecture of our proposed M2FNet is shown in Fig. 1. The framework mainly consists of two branches, namely, the HSI branch and the LiDAR data branch. First, a pair of HSI and LiDAR data covering the same area of the Earth's surface are fed into two branched feature extraction backbones to extract spatial, spectral, and elevation features of land cover objects, respectively. Then, in the DConvformer module, the extracted features first pass through an FE module, and then the enhanced features are sent into the transformer structure. In this structure, we use dilated CNN network layers to replace the self-attention module in the traditional transformer and deeply integrate heterogeneous information. Finally, the two outputs of DConvformer are passed into the classifier head to get the pixel-level classification result.

B. HSI and LiDAR Data Preprocessing

Consider a set of HSI data, denoted by $\boldsymbol{X}_{\mathrm{HSI}} \in \mathbb{R}^{m\times n\times b}$ and a set of LiDAR data covering the same area on the Earth's surface, denoted by $\boldsymbol{X}_{\mathrm{LiDAR}} \in \mathbb{R}^{m\times n}$, where m and n represent the two spatial sizes and b is the number of HSI spectral bands. For each pixel, we extract a 3-D cubic patch $\boldsymbol{X}_p^H \in \mathbb{R}^{s\times s\times b}$ and a 2-D square patch $\boldsymbol{X}_p^L \in \mathbb{R}^{s\times s}$ from HSI and LiDAR data, respectively. Here, $(s \times s)$ represents the patch size. The category of each pixel can be represented by a one-hot category vector.

After extraction, all the cubic patches $\boldsymbol{X}_p^H$ from the HSI data are collected to form a set $\boldsymbol{\Gamma}_H$. Similarly, for LiDAR data, all the corresponding square patches $\boldsymbol{X}_p^L$ are collected to form a set $\boldsymbol{\Gamma}_L$. Finally, the training and test sets can be generated by the following equations:

$$\mathcal{D}^{\mathrm{train}} = \left\{ \left(\boldsymbol{x}_p^h, \boldsymbol{x}_p^l\right)^{(i)}, \boldsymbol{y}^{(i)} | i = 1, \ldots, n_1 \right\} \tag{1}$$

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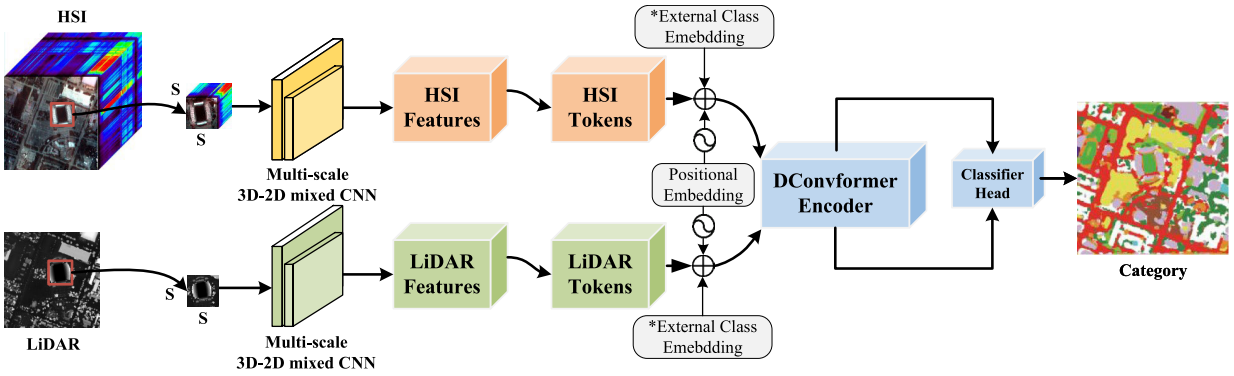


Fig. 1. Overall architecture of the proposed network for HSI and LiDAR fusion and classification.

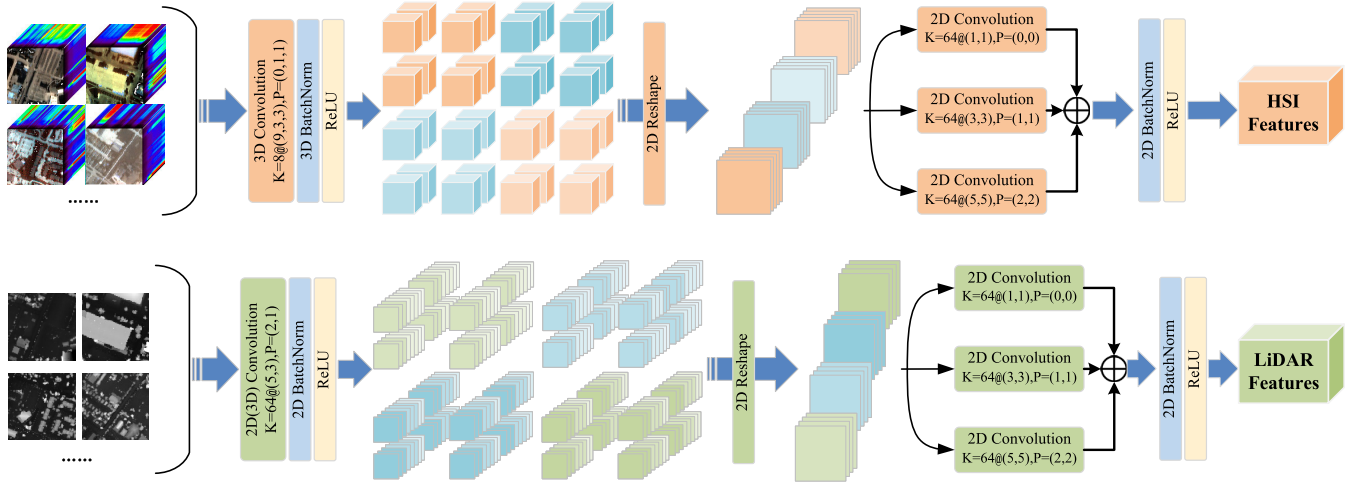


Fig. 2. Feature learning via multiscale 3-D–2-D mixed CNN. In convolutional operations, "K" represents the size of the convolutional kernel, and "P" denotes the padding size.

$$\mathcal{D}^{\text{test}} = \left\{ \left(\boldsymbol{x}_p^h, \boldsymbol{x}_p^l \right)^{(j)}, \boldsymbol{y}^{(j)} | j = 1, \ldots, n_2, j \neq i \right\} \quad (2)$$

where $\boldsymbol{x}_p^h \in \boldsymbol{\Gamma}_H$ and $\boldsymbol{x}_p^l \in \boldsymbol{\Gamma}_L$ are randomly chosen from the two patch sets, n_1 and n_2 represent the number of training and test samples, respectively, and $\boldsymbol{y}^{(i)}$ is the corresponding category vector, and it is extracted from the ground-truth data $\boldsymbol{Y} \in \mathbb{R}^{m \times n}$.

C. HSI and LiDAR Feature Learning via Multiscale 3-D–2-D Mixed CNN

CNNs can gradually decrease the spatial dimension of input data using multilayer convolution and pooling operations, enabling them to model images at various scales and spatial levels. By leveraging their exceptional capability to model spatial contextual features, the combination of CNN with HSI classification can comprehensively exploit the benefits of CNN in image processing. This approach enables effective handling of spatial correlations in HSI data, thorough extraction of spectral information from HSI data, and extraction of elevation information from LiDAR data. Consequently, it enhances classification accuracy and stability. Our proposed model uses a multiscale 3-D–2-D mixed CNN to extract deep-level abstract features.

Fig. 2 illustrates the utilization of sequential Conv3-D and multiscale Conv2-D layers in our proposed model for extracting discriminative spatial and spectral features from HSI data. The size of each HSI patch cube $\boldsymbol{X}_p^H$ in the training set is $s \times s \times b$. It is used as input training data for the Conv3-D layer. In the Conv3-D layer, the size of the convolutional kernel is set to $8@3 \times 3 \times 9$ and the size of the padding is $(1 \times 1 \times 0)$. We apply padding to ensure the output image has the same dimensions as the input image. The multiscale Conv2-D module consists of three parallel Conv2-D layers after the Conv3-D layer. Two Conv2-D layers are used for grouping convolution, and the other is used for point-by-point convolution. This module uses three kernels of different sizes to extract information at multiple scales. The features $\boldsymbol{F}^H$ obtained from these three convolutions are combined in an elemental manner ($\oplus$)

$$\hat{\boldsymbol{X}}_p^H = \text{Reshape}\left(\text{Conv3D}\left(\boldsymbol{X}_p^H\right)\right) \quad (3)$$

$$\begin{aligned} \boldsymbol{F}^H = & Conv2D\left(\hat{\boldsymbol{X}}_p^H, k_1, g_1, p_1\right) \\ & \oplus Conv2D\left(\hat{\boldsymbol{X}}_p^H, k_2, g_2, p_2\right) \\ & \oplus Conv2D\left(\hat{\boldsymbol{X}}_p^H, k_3, g_3, p_3\right) \end{aligned} \quad (4)$$

where $k_1 = 1$, $g_1 = 1$, $p_1 = 0$, $k_2 = 3$, $g_2 = 4$, $p_2 = 1$, $k_3 = 5$, $g_3 = 4$, and $p_3 = 2$.

After performing the convolution operation, the 3-D features of HSIs can be obtained. In contrast to HSI processing, when

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LiDAR data contain multiple channels, we use Conv3-D to extract more effective and prominent spatial and elevation features. As depicted in Fig. 2, when there is only one channel, we use the more appropriate Conv2-D to extract data features. Taking Conv2-D as an example, the size of each LiDAR patch $\boldsymbol{X}_p^L$ is $s \times s$. This patch is used as the input for the Conv2-D layer. In the Conv2-D layer, the size of the convolutional kernel is set to $64@5 \times 3$ and the fill size is set to (2×1). In the multiscale Conv2-D module, we adopt a network structure similar to the HSI branch to enhance feature extraction. Therefore, the output $\boldsymbol{F}^L$ of this module in the second branch can be expressed by the following equation:

$$\hat{\boldsymbol{X}}_p^L = \text{Reshape}\big(\text{Conv2D}\big(\boldsymbol{X}_p^L\big)\big) \tag{5}$$

$$\begin{aligned}\boldsymbol{F}^L = &\text{Conv2D}\Big(\hat{\boldsymbol{X}}_p^L, k_1, g_1, p_1\Big) \\ &\oplus \text{Conv2D}\Big(\hat{\boldsymbol{X}}_p^L, k_2, g_2, p_2\Big) \\ &\oplus \text{Conv2D}\Big(\hat{\boldsymbol{X}}_p^L, k_3, g_3, p_3\Big)\end{aligned} \tag{6}$$

where $k_1 = 1$, $g_1 = 1$, $p_1 = 0$, $k_2 = 3$, $g_2 = 4$, $p_2 = 1$, $k_3 = 5$, $g_3 = 4$, and $p_3 = 2$. To solve the problem of merging and accelerating training performance, batch normalization (BN) and ReLU activation layers are used after the convolutional layer; here, ReLU helps smooth the backpropagation of the loss by introducing nonlinearity.

D. Feature Tokenization and Position Embedding

The spatial–spectral features of HSI and the elevation features of LiDAR data, extracted by the double-branch multiscale convolutional network, contain abundant discriminative information, which can greatly improve classification accuracy. Next, these features need to be tokenized and embedded into the transformer network for further processing.

The flattened HSI and LiDAR data feature maps are represented by $\boldsymbol{F}_{\text{flat}}^H \in \mathbb{R}^{u_h v_h \times z_h}$ and $\boldsymbol{F}_{flat}^L \in \mathbb{R}^{u_l v_l \times z_l}$, respectively, where u_h and v_h are the height and width of HSI feature maps, respectively, and u_l and v_l are the height and width of LiDAR data feature maps respectively. z_h and z_l are the number of feature channels of HSI and LiDAR data feature maps, respectively. The above procedure can be expressed as

$$\boldsymbol{F}_{\text{flat}}^H = \boldsymbol{\mathcal{TF}}\Big(\text{Flatten}\Big(\hat{\boldsymbol{F}}^H\Big)\Big) \tag{7}$$

$$\boldsymbol{F}_{\text{flat}}^L = \boldsymbol{\mathcal{TF}}\Big(\text{Flatten}\Big(\hat{\boldsymbol{F}}^L\Big)\Big) \tag{8}$$

where $\boldsymbol{\mathcal{TF}}(\cdot)$ is a transpose function, and $\hat{\boldsymbol{F}}^H$ and $\hat{\boldsymbol{F}}^L$ are the results obtained after BN and ReLU activation layers for $\boldsymbol{F}^H$ and $\boldsymbol{F}^L$, respectively. The tokenization process uses two learnable weights $\boldsymbol{W}_A$ and $\boldsymbol{W}_B$, and intermediate variables $\boldsymbol{P}^H$ and $\boldsymbol{P}^L$ can be obtained by the following equations:

$$\boldsymbol{P}^H = \text{softmax}\big(\boldsymbol{\mathcal{TF}}\big(\boldsymbol{F}_{\text{flat}}^H \boldsymbol{W}_A\big)\big) \tag{9}$$

$$\boldsymbol{P}^L = \text{softmax}\big(\boldsymbol{\mathcal{TF}}\big(\boldsymbol{F}_{\text{flat}}^L \boldsymbol{W}_B\big)\big). \tag{10}$$

Here, $\boldsymbol{F}_{\text{flat}}^H \boldsymbol{W}_A$ and $\boldsymbol{F}_{\text{flat}}^L \boldsymbol{W}_B$ are 1×1 dot products. HSI feature tokens are represented by $\boldsymbol{T}_{\text{patch}}^H \in \mathbb{R}^{w_h \times z_h}$, where w_h is the number of HSI tokens. LiDAR data feature tokens are represented as $\boldsymbol{T}_{\text{patch}}^L \in \mathbb{R}^{w_l \times z_l}$, where w_l is the number of LiDAR data tokens. Given the features $\boldsymbol{F}_{\text{flat}}^H$ and $\boldsymbol{F}_{\text{flat}}^L$, $\boldsymbol{T}_{\text{patch}}^H$ and $\boldsymbol{T}_{\text{patch}}^L$ can be obtained by the following equation:

$$\boldsymbol{T}_{\text{patch}}^H = \boldsymbol{P}^H \boldsymbol{F}_{\text{flat}}^H \tag{11}$$

$$\boldsymbol{T}_{\text{patch}}^L = \boldsymbol{P}^L \boldsymbol{F}_{\text{flat}}^L. \tag{12}$$

All the tokens are obtained by concatenating ($\odot$) an all-zero learnable classification token $\boldsymbol{T}_{\text{cls}}^H \in \mathbb{R}^{1 \times z_h}$ to the HSI patch tokens $\boldsymbol{T}_{\text{patch}}^H$. The processing of LiDAR data follows a similar procedure. Semantic texture information in image block tags can be retained by adding trainable positional embeddings into block embeddings. Consequently, trainable positional embeddings are added to the generated HSI tokens and LiDAR tokens. Here, we use a dropout layer after this operation to mitigate the impact of the vanishing gradient. The aforementioned procedure is expressed as

$$\boldsymbol{T}_{\text{tokens}}^H = \boldsymbol{DP}\Big[\Big(\boldsymbol{T}_{\text{patch}}^H \odot \boldsymbol{T}_{\text{cls}}^H\Big) \oplus \boldsymbol{PE}_H\Big] \tag{13}$$

$$\boldsymbol{T}_{\text{tokens}}^L = \boldsymbol{DP}\Big[\Big(\boldsymbol{T}_{\text{patch}}^L \odot \boldsymbol{T}_{\text{cls}}^L\Big) \oplus \boldsymbol{PE}_L\Big] \tag{14}$$

where $\boldsymbol{DP}$ represents a dropout layer with a value of 0.1 and $\boldsymbol{PE}$ represents a learnable positional embedding.

E. DConvformer Encoder

The importance of feature fusion in multisource remote sensing data classification cannot be underestimated. By fully using the information from multiple sources, feature fusion can enhance the accuracy of classification, improve the robustness of the model, and support the analysis and monitoring of complex surface scenes. However, integrating feature information from different levels and types remains a challenge. To address this issue, we have carefully designed the DConvformer encoder to integrate features from HSI and LiDAR data. Specifically, we first input the feature embeddings of both data types, excluding the learnable classification token, into the morphological FE module. This module mainly consists of a spatial morphological (SpatialMorph) block and a spectral morphological (SpectralMorph) block. The enhanced features will serve as high-quality inputs for the subsequent component. To effectively fuse the features of these two heterogeneous data types, the respective learnable classification tokens of the two data are exchanged after feature learning using the deep dilated convolution module. Interacting the learnable classification token, which has already learned abstract information from all its feature tokens, with feature tokens from the other data mode is beneficial for acquiring supplementary information. The unfolded DConvformer encoder is illustrated in Fig. 3.

1) Morphological FE Module: Mathematical morphology is a powerful image processing technique based on set theory and algebraic operations. It involves applying operations such as erosion, dilation, opening, and closing to alter the shape and structure of objects in an image. Erosion operation can shrink or eliminate objects in an image, while dilation operation can expand or fill the holes in objects. As shown in the unfolded DConvformer encoder, it is designed based on dilation and erosion operations using a structural element of size $(r \times r)$.

In this network, the erosion operation involves subtracting a kernel from the local region of an image to select the minimum

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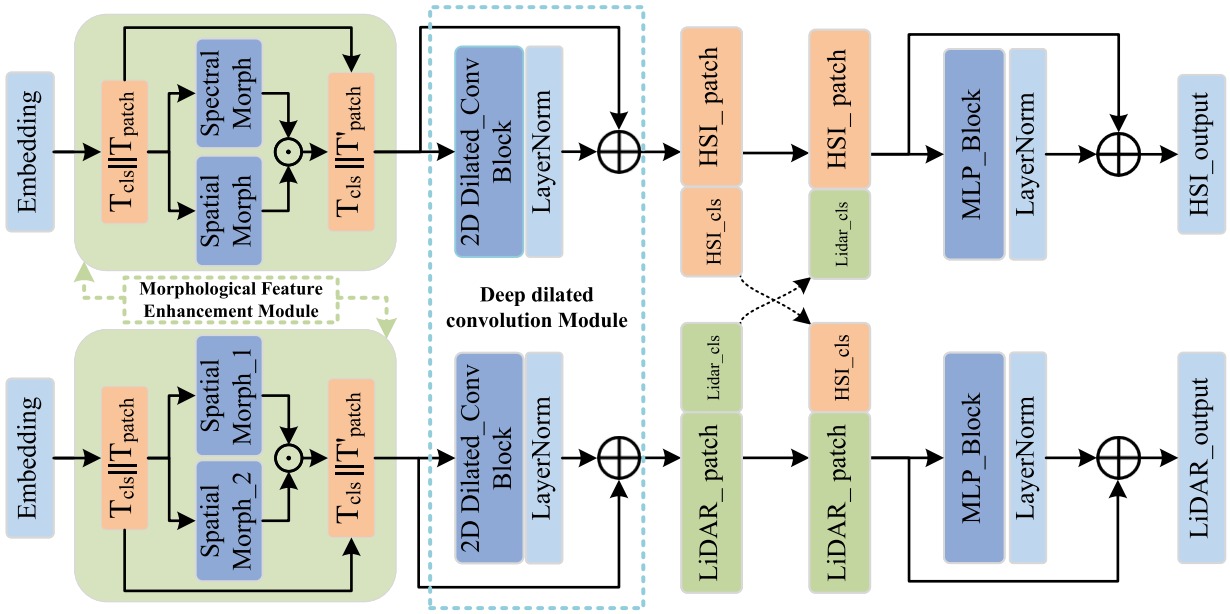


Fig. 3. Architecture of the unfolded DConvformer encoder.

value within the current operation area. This helps reduce the shape of background objects in the tokens, making them easier to distinguish in different backgrounds. Let $\boxminus$ represent the morphological erosion operation, the erosion operation can be defined as

$$(\boldsymbol{Q} \boxminus \boldsymbol{W}_e)(x, y) = \min_{(a_1,a_2)\in\boldsymbol{\varphi}} (\boldsymbol{Q}(x + a_1, y + a_2) - \boldsymbol{W}_e(a_1, a_2)) \quad (15)$$

where $\boldsymbol{\varphi} = \{(a_1, a_2)|a_1 \in \{1, 2, \ldots, r\}; a_2 \in \{1, 2, \ldots, r\}\}$ represents the elements of the kernel, $\boldsymbol{Q}$ represents the feature data from which morphological information is to be extracted, and $\boldsymbol{W}_e$ denotes the structural elements used for the erosion operation.

Similar to the erosion operation, the dilation operation involves applying a kernel to the local region of an image. However, in contrast, this operation is based on addition and selects the pixel with the maximum value within the local neighborhood. This operation expands the boundaries of foreground objects within the tokens. The dilation process is represented by $\boxplus$ and can be denoted by the following equation:

$$(\boldsymbol{Q} \boxplus \boldsymbol{W}_d)(x, y) = \max_{(a_1,a_2)\in\boldsymbol{\varphi}} (\boldsymbol{Q}(x + a_1, y + a_2) + \boldsymbol{W}_d(a_1, a_2)) \quad (16)$$

where $\boldsymbol{W}_d$ denotes the structural elements used for the dilation operation.

As shown in Fig. 4, we used a 3×3-sized structural element. In (a), the erosion operation is demonstrated, while (b) illustrates the dilation operation. To fully extract SpatialMorph information embedded in HSI, we performed these two operations in parallel and then merged them to form the SpatialMorph module. To prevent feature variability, we introduced convolutional operations and performed normalization after fusion. The entire SpatialMorph block can be described as

$$\boldsymbol{T}^H_{\text{SpatMorph}} = \mathcal{F}^d_{2D_{(3\times3)}}\left(\boldsymbol{T}^H_{\text{patch}} \boxminus \boldsymbol{W}_e\right) \oplus \mathcal{F}^e_{2D_{(3\times3)}}\left(\boldsymbol{T}^H_{\text{patch}} \boxplus \boldsymbol{W}_d\right) \quad (17)$$

where $\boldsymbol{W}_d$ and $\boldsymbol{W}_e$ are the weights of the (3×3) convolutional kernel, and $\mathcal{F}^d_{2D}$ and $\mathcal{F}^e_{2D}$, respectively, represent the linear combinations of Conv2-D with dilation and erosion operations, as illustrated in Fig. 5. The kernel size of the 2-D convolutional layer is 3×3. To extract SpectralMorph features from HSI patch tokens, a SpectralMorph module similar to SpatialMorph is used, but with a convolutional kernel size of 1×1. This block can be described using the following equation:

$$\boldsymbol{T}^H_{\text{SpecMorph}} = \mathcal{F}^d_{2D_{(1\times1)}}\left(\boldsymbol{T}^H_{\text{patch}} \boxminus \boldsymbol{W}_e\right) \oplus \mathcal{F}^e_{2D_{(1\times1)}}\left(\boldsymbol{T}^H_{\text{patch}} \boxplus \boldsymbol{W}_d\right). \quad (18)$$

The LiDAR data branch adopts a similar structure and uses two SpatialMorph blocks with different convolutional kernel sizes in parallel

$$\boldsymbol{T}^L_{\text{SpatMorph}_1} = \mathcal{F}^d_{2D_{(3\times3)}}\left(\boldsymbol{T}^L_{\text{patch}} \boxminus \boldsymbol{W}_e\right) \oplus \mathcal{F}^e_{2D_{(3\times3)}}\left(\boldsymbol{T}^L_{\text{patch}} \boxplus \boldsymbol{W}_d\right) \quad (19)$$

$$\boldsymbol{T}^L_{\text{SpatMorph}_2} = \mathcal{F}^d_{2D_{(1\times1)}}\left(\boldsymbol{T}^L_{\text{patch}} \boxminus \boldsymbol{W}_e\right) \oplus \mathcal{F}^e_{2D_{(1\times1)}}\left(\boldsymbol{T}^L_{\text{patch}} \boxplus \boldsymbol{W}_d\right). \quad (20)$$

In conclusion, the morphological FE module can be summarized by the following formula:

$$\boldsymbol{T}^H_{FE} = \boldsymbol{T}^H_{\text{SpatMorph}} \odot \boldsymbol{T}^H_{\text{SpecMorph}} \odot \boldsymbol{T}^H_{\text{cls}} \quad (21)$$

$$\boldsymbol{T}^L_{FE} = \boldsymbol{T}^L_{\text{SpatMorph}_1} \odot \boldsymbol{T}^L_{\text{SpatMorph}_2} \odot \boldsymbol{T}^L_{\text{cls}}. \quad (22)$$

Here, $\boldsymbol{T}^H_{FE}$ and $\boldsymbol{T}^L_{FE}$ represent the outputs of the two branches passing through this module, while $\boldsymbol{T}^H_{\text{cls}}$ and $\boldsymbol{T}^L_{\text{cls}}$ are the learnable classification tokens in (13) and (14), respectively.

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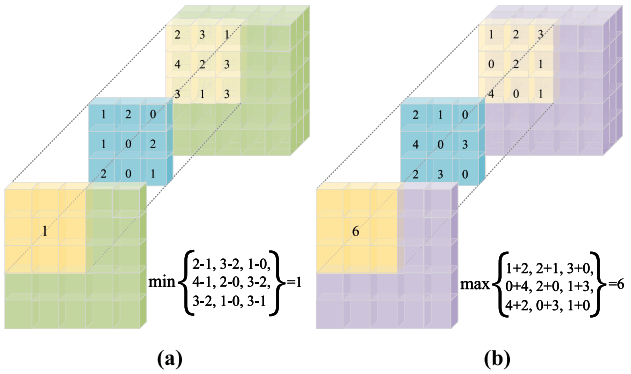


Fig. 4. Graphical visualization of (a) erosion operation and (b) dilation operation.

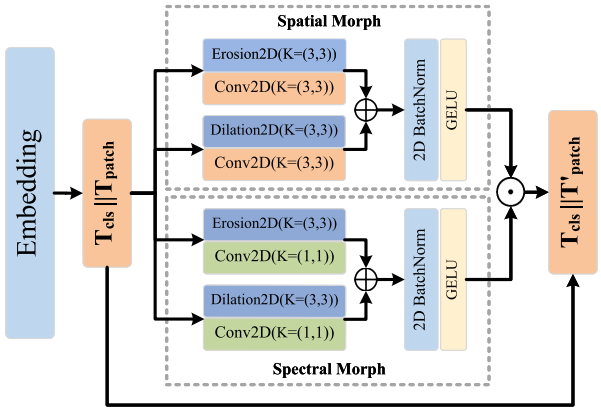


Fig. 5. Graphical illustration of the FE module. In convolutional operations, "K" represents the size of the convolutional kernel.

2) Deep Dilated Convolution Module: Compared with the powerful CNN network, the attention mechanism has some problems such as high computational complexity and lack of local information. To solve this problem, the proposed method uses a deep dilated convolutional module to replace the self-attention module. This module uses six dilated Conv2-D layers, each with a kernel size of (3×3), and dilation rates of 1, 2, 5, 1, 2, and 5, respectively. The number of output channels for each layer is set as 8, 16, 32, 16, 8, and 1. The different dilation rate makes the receptive field different and can capture multiscale context information. $\boldsymbol{T}^H_{\text{DConv}}$ and $\boldsymbol{T}^L_{\text{DConv}}$ are obtained from $\boldsymbol{T}^H_{FE}$ and $\boldsymbol{T}^L_{FE}$, respectively. $\boldsymbol{T}^H_{FE}$ and $\boldsymbol{T}^L_{FE}$ are the outputs of the FE module. The outputs of this module are $\boldsymbol{T}^H_{\text{DConv}}$ and $\boldsymbol{T}^L_{\text{DConv}}$, which can be calculated using the following formula:

$$\boldsymbol{T}^H_{\text{DConv}} = \boldsymbol{LN}\left[\boldsymbol{\mathcal{F}}_{\text{DConv2D}}\left(\boldsymbol{T}^H_{FE}\right)\right] \oplus \boldsymbol{T}^H_{FE} \tag{23}$$

$$\boldsymbol{T}^L_{\text{DConv}} = \boldsymbol{LN}\left[\boldsymbol{\mathcal{F}}_{\text{DConv2D}}\left(\boldsymbol{T}^L_{FE}\right)\right] \oplus \boldsymbol{T}^L_{FE}. \tag{24}$$

$\boldsymbol{\mathcal{F}}_{\text{DConv2D}}$ is the concatenation function of dilated Conv2-D. $\boldsymbol{LN}$ stands for layer normalization. In the transformer architecture, it is used to normalize each feature dimension, thus maintaining a stable distribution for each feature during training. Next, we interchange the respective learnable classification tokens of the two branches to fully integrate the heterogeneous information. The interchanged $\boldsymbol{T}^H_{\text{swap}}$ and $\boldsymbol{T}^L_{\text{swap}}$ are obtained by the following formula:

$$\boldsymbol{T}^H_{\text{swap}} = \hat{\boldsymbol{T}}^L_{\text{cls}} \odot \hat{\boldsymbol{T}}^H_{\text{patch}} \tag{25}$$

$$\boldsymbol{T}^L_{\text{swap}} = \hat{\boldsymbol{T}}^H_{\text{cls}} \odot \hat{\boldsymbol{T}}^L_{\text{patch}} \tag{26}$$

where $\hat{\boldsymbol{T}}^L_{cls}, \hat{\boldsymbol{T}}^H_{\text{patch}}, \hat{\boldsymbol{T}}^H_{\text{cls}}, \hat{\boldsymbol{T}}^L_{\text{patch}}$ are extracted from the output of the previous component, i.e., $\boldsymbol{T}^H_{\text{DConv}} = \hat{\boldsymbol{T}}^H_{\text{cls}} \odot \hat{\boldsymbol{T}}^H_{\text{patch}}$ and $\boldsymbol{T}^L_{\text{DConv}} = \hat{\boldsymbol{T}}^L_{\text{cls}} \odot \hat{\boldsymbol{T}}^L_{\text{patch}}$. Finally, the outputs of DConvformer can be represented by the following formula:

$$\boldsymbol{D}^H_{\text{out}} = \boldsymbol{LN}\left[\boldsymbol{MLP}\left(\boldsymbol{T}^H_{\text{swap}}\right)\right] \oplus \boldsymbol{T}^H_{\text{swap}} \tag{27}$$

$$\boldsymbol{D}^L_{\text{out}} = \boldsymbol{LN}\left[\boldsymbol{MLP}\left(\boldsymbol{T}^L_{\text{swap}}\right)\right] \oplus \boldsymbol{T}^L_{\text{swap}} \tag{28}$$

where $\boldsymbol{MLP}(\cdot)$ is the multilayer perceptron. The MLP includes two linear layers with Gaussian error linear unit (GELU) operations.

F. Classifier Head

The two learnable classification tokens from $\boldsymbol{D}^H_{\text{out}}$ and $\boldsymbol{D}^L_{\text{out}}$ output by DConvformer are, respectively, fed to the MLP head, and then their outputs are fused to get the output of the whole network. The MLP head includes the layer norm layer and linear layer, where the last linear layer with softmax function aims to generate the final label for joint classification. In MLP head, the size of the output layer is set to be the same as the total number of classes. The above process can be represented by the following formula:

$$\boldsymbol{Out} = \boldsymbol{MH}\left[\boldsymbol{D}^{\text{Hout}}_{\text{cls}}\right] \oplus \boldsymbol{MH}\left[\boldsymbol{D}^{\text{Lout}}_{cls}\right] \tag{29}$$

where $\boldsymbol{MH}[\cdot]$ represents the MLP head, and $\boldsymbol{D}^{\text{Hout}}_{\text{cls}}$ and $\boldsymbol{D}^{\text{Lout}}_{\text{cls}}$ are learnable classification tokens extracted from $\boldsymbol{D}^H_{out}$ and $\boldsymbol{D}^L_{\text{out}}$, respectively. Softmax ensures that the total activation of each output unit is 1. Therefore, the output can be considered as a set of conditional probabilities. Then, the last two output probability vectors are added to obtain the final probability vector. The label corresponding to the maximum probability is the category of pixels. The process of obtaining the category label can be expressed by the following equation:

$$\text{Label} = \max\left(\underbrace{\text{softmax}(\boldsymbol{Out})}_{\boldsymbol{I}}\right) \tag{30}$$

where $\boldsymbol{I} \in \mathbb{R}^{1\times C}$, and C represents the number of categories. The max function is used to obtain the maximum value in this variable to generate the category label corresponding to the pixel. The cross-entropy loss function is used as the loss function in the integrated network, as shown in the following formula:

$$H(p, q) = -\sum_{k=1}^{C} p(x_k) \log q(x_k) \tag{31}$$

where $p(x_k)$ is the predictive probability that the pixel belongs to class k. $q(x_k)$ is the expected probability that the pixel belongs to class k.

The overall training process of the proposed method is shown in Algorithm 1.

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Algorithm 1 Multiscale Feature Extraction and Multisource Data Fusion Model

Input: Input an HSI data $\boldsymbol{X}_{HSI} \in \mathbb{R}^{m\times n\times b}$, an LiDAR data $\boldsymbol{X}_{LiDAR} \in \mathbb{R}^{m\times n}$ and a ground-truth data $\boldsymbol{Y} \in \mathbb{R}^{m\times n}$; patch size $s = 11$; training sample rate $\delta\%$.

Output: Predicted labels of the test set.

Set batch size to 64, optimizer Adam (learning rate: $5e-4$), learning rate scheduler StepLR (step_size=50, $\gamma = 0.9$), epochs number ϵ to 500, initialize all weights.
Create all sample patches from $\boldsymbol{X}_{HSI}$ and $\boldsymbol{X}_{LiDAR}$, and divide them into a training set and a test set.
Generate training loader and test loader.
for $i = 1$ to ϵ **do**
    Use multiscale 3-D–2-D mixed CNN module to extract shallow spatial–spectral features $\boldsymbol{F}^H$ from $\boldsymbol{X}_p^H$, and extract shallow elevation features from $\boldsymbol{X}_p^L$ to obtain $\boldsymbol{F}^L$.
    Convert the obtained shallow 2-D feature maps into feature tokens and incorporate them with learnable classification tokens and positional information, as described by(7)–(14).
    Input the feature tokens into the DConvformer encoder for deep feature learning and fusion.
    Perform the DConvformer encoder by (21)–(28).
    Input the learnable classification tokens from $\boldsymbol{D}_{out}^H$ and $\boldsymbol{D}_{out}^L$ to the MLP head and fuse them.
    Obtain the label corresponding to the current pixel using (30).
end for
Use the test set with the trained model to get predicted labels.

III. Experiments and Analysis

Three publicly accessible multimodal remote sensing datasets were used for experiments to assess the effectiveness of the proposed framework. First, detailed information regarding the three datasets used in the experiments was presented. Subsequently, we described the specifics of the experimental settings and showcased the experimental results, which demonstrated the effectiveness of the proposed method and its superiority over the existing approaches. Finally, ablation experiments were conducted to illustrate the functionality of the various modules within the proposed framework.

A. Data Descriptions

To evaluate the effectiveness of the proposed framework, The Houston2013 dataset, Trento dataset, and Augsburg dataset are used for quantitative and qualitative performance assessment.

1) Houston2013 Dataset: The Houston2013 dataset was provided by the 2013 IEEE GRSS Data Fusion Contest. The dataset covers the University of Houston campus and the surrounding urban area was collected by the NSF-funded National Center for Airborne Laser Mapping in 2012. For the same region, LiDAR data were provided with only one band. Both the images contain 349 × 1905 pixels, and the HSI contains 144 spectral bands ranging from 0.38 to 1.05 μm. This dataset depicts 15 urban land-cover classes. Fig. 6 visualizes a pseudocolor composite image of the HSI data, a grayscale image of the LiDAR data, and the ground-truth map.

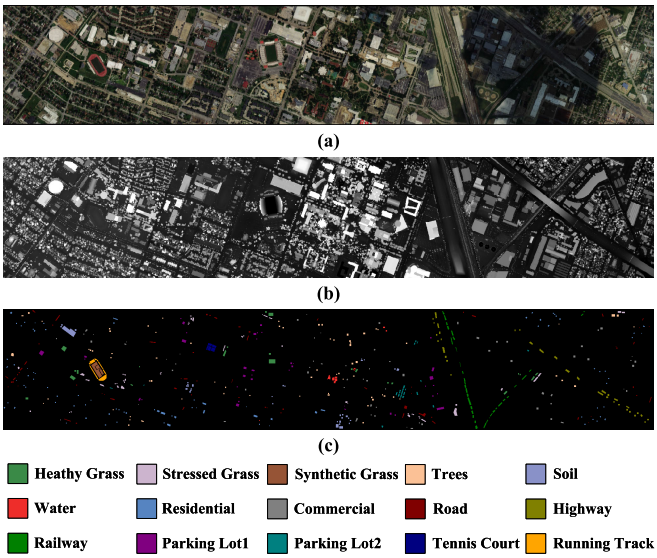


Fig. 6. Houston2013 dataset. (a) Pseudocolor composite image based on bands 64, 43, and 20 for HSIs. (b) Grayscale image for LiDAR-based DSM. (c) Ground truth.

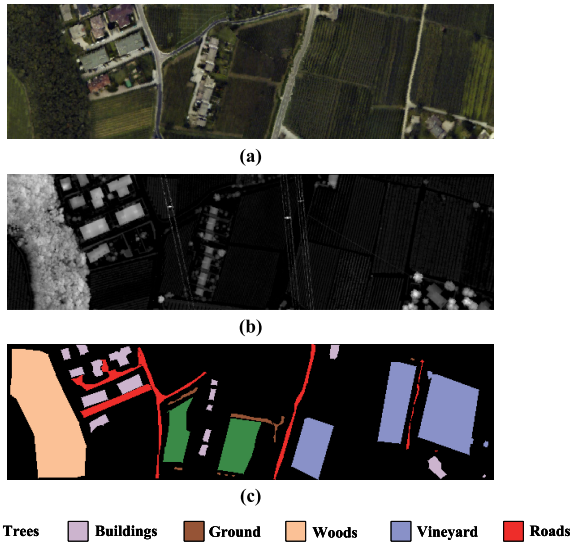


Fig. 7. Trento dataset. (a) Pseudocolor composite image based on bands 20, 15, and 5 for HSIs. (b) Grayscale image for LiDAR-based DSM. (c) Ground truth.

2) Trento Dataset: The Trento dataset was captured over a rural region in the south of Trento, Italy. The HSI data were obtained by the AISA Eagle system for hyperspectral imaging Eagle sensor with 63 spectral bands and a spectral resolution ranging from 0.42 to 0.99 μm, and the LiDAR data were collected by the Optech ALTM 3100EA sensor with 1 raster. The size of this dataset is 600 × 166 pixels with a spatial resolution of 1 m and includes six different types of ground objects. Fig. 7 shows a pseudocolor composite image of the HSI data, a grayscale image of the LiDAR data, and the ground-truth map, respectively.

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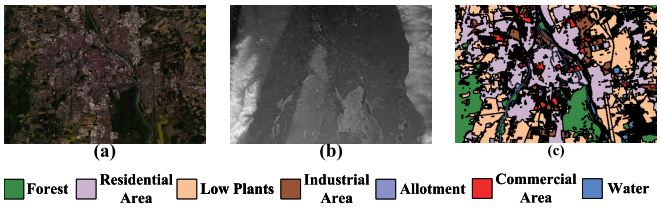


Fig. 8. Augsburg dataset. (a) Pseudocolor composite image based on bands 20, 15, and 5 for HSIs. (b) Grayscale image for the LiDAR-based DSM. (c) Ground truth.

3) Augsburg Dataset: The Augsburg dataset was captured over the city of Augsburg, Germany. The HSI data were acquired by the DAS-EOC HySpex sensor, and the LiDAR-based DSM data were collected by the DLR-3K system. The spatial resolutions of both the images were downsampled to a unified resolution of 30 m to adequately manage the multimodal fusion. In this dataset, the HSI data consist of 180 bands ranging from 0.4 to 2.5 μm, while the DSM data have a single raster. The size of this dataset is 332 $\times$ 485 pixels. The dataset depicts seven different land- cover classes. In Fig. 8, the first image illustrates a pseudocolor composite image of the HSI data, followed by a grayscale image of the LiDAR data, and finally, the ground-truth map is displayed.

The names of the land-cover categories, the number of training samples, and the number of test samples for these three datasets are listed in Table I.

B. Experimental Settings

1) Evaluation Indicators: To evaluate the classification performance of the proposed framework and other existing models, four widely used quantity analysis criteria were computed, namely, the overall accuracy (OA), average accuracy (AA), kappa coefficient (κ), and per-class accuracy. For each indicator, a higher value indicates a more accurate classification accuracy. The experiments have been performed on spectrally and spatially disjoint sets of training and test samples such that there is no interaction between the respective samples.

2) Configurations: The experiments with the proposed method and other deep learning methods were all implemented in the PyTorch platform, using a server with an Intel Xeon Silver 4314 2.40-GHz CPU, 192-GB RAM, and two NVIDIA GeForce RTX 4090 GPUs with 24 GB of VRAM. To optimize the network, the Adam optimizer was selected as the initial optimizer. For the training stage, the batch size was set to 64. In addition, we used a step scheduler with a gamma of 0.9 and step size of 50 and conducted training over 500 epochs. The average and standard deviation of each experiment have been calculated based on three repetitions. Python 3.8.16 and PyTorch 2.0.0 were used to implement the coding of the proposed M2FNet.

3) Supplementary Details: To fully demonstrate the generalization ability of M2FNet, we have supplemented the Supplementary Material with experiments on the Houston2018 dataset and provided detailed explanations.

C. Parameter Analysis

1) Batch Size: The batch size affects the network in terms of training speed, memory usage, accuracy, and generalization capability. A larger batch size can accelerate training speed but may consume more memory. On the other hand, a smaller batch size might lead to overfitting and be susceptible to training noise. Choosing an appropriate batch size requires considering the specific problem and hardware setup, and finding the optimal value through experimentation. To evaluate the effect of the batch size, the other hyperparameter values were fixed and batch size was selected from a candidate set {16, 32, 64, 128, 256}. It is evident from Fig. 9(a) that batch size has an effect on OA. When the batch size is set to 64, our proposed method achieves the optimal solution across all the three datasets.

2) Patch Size: The patch size in HSI classification has an impact on the network. The choice of patch size determines the spatial context considered by the network during classification. A larger patch size captures more contextual information but increases computational complexity and memory requirements. Conversely, a smaller patch size focuses on local details but may overlook important spatial relationships. The selection of patch size should be carefully balanced to ensure optimal performance and efficiency in hyperspectral classification tasks. Similar to the analysis of batch size, the other hyperparameters were fixed. We selected the patch size from a candidate set {7, 9, 11, 13, 15} to evaluate its effect. Fig. 9(b) illustrates the impact of different patch sizes on OA. When the patch size was set to 11, the optimal OA value was achieved across all the three datasets.

3) Learning Rate: Fig. 9(c) shows the influence of the learning rate on the OA of the three datasets. The learning rate greatly affects the performance and accuracy of deep neural networks. A higher learning rate leads to faster convergence initially but risks overshooting or divergence. A lower learning rate provides more precise updates and potentially higher accuracy but slows down training. Finding the right learning rate requires experimentation and balancing convergence speed with accuracy improvement. In experiments, the learning rate was selected from a candidate set {1e–4, 3e–4, 5e–4, 1e–3, 5e–3}. When the learning rate was set to 5e$-$4, it was obvious that we obtained the optimal solution of OA.

4) Step Size: The step size has a significant impact on the performance and accuracy of deep networks in step scheduler. A smaller step size can accelerate convergence, capture subtle variations, and improve performance, but it may lead to unstable training. On the other hand, a larger step size can maintain stability but may result in slower convergence and decreased accuracy. Choosing an appropriate step size requires balancing computational efficiency and performance, adjusting based on network architecture, data characteristics, and task requirements, and progressively optimizing for better results. The step size was selected from a candidate set {10, 50, 100, 130, 160, 200}. Fig. 10 shows the changes in OA in the relationship between different learning rates and step sizes. We observed that when the learning rate was set to 5e$-$4 and the step size was set to 50, the OA consistently achieved the best results on all the three datasets.

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TABLE I
TRAINING AND TEST SAMPLE NUMBERS FOR THE HOUSTON2013, TRENTO, AND AUGSBURG DATASETS

NO.	Houston2013 dataset			Trento dataset			Augsburg dataset		
	Class Name	Training.	Test.	Class Name	Training.	Test.	Class Name	Training.	Test.
C01	Healthy Grass	198	1053	Apple Trees	40	3994	Forest	675	12832
C02	Stressed Grass	190	1064	Buildings	29	2874	Residential Area	1516	28813
C03	Synthetic Grass	192	505	Ground	5	474	Industrial Area	192	3659
C04	Trees	188	1056	Woods	91	9032	Low Plants	1342	25515
C05	Soil	186	1056	Vineyard	105	10396	Allotment	28	547
C06	Water	182	143	Roads	31	3143	Commercial Area	82	1563
C07	Residential	196	1072				Water	76	1454
C08	Commercial	191	1053						
C09	Road	193	1059						
C10	Highway	191	1036						
C11	Railway	181	1054						
C12	Parking Lot1	192	1041						
C13	Parking Lot2	184	285						
C14	Tennis Court	181	247						
C15	Running Track	187	473						
-	Total	2832	12197	Total	301	29913	Total	3911	74383

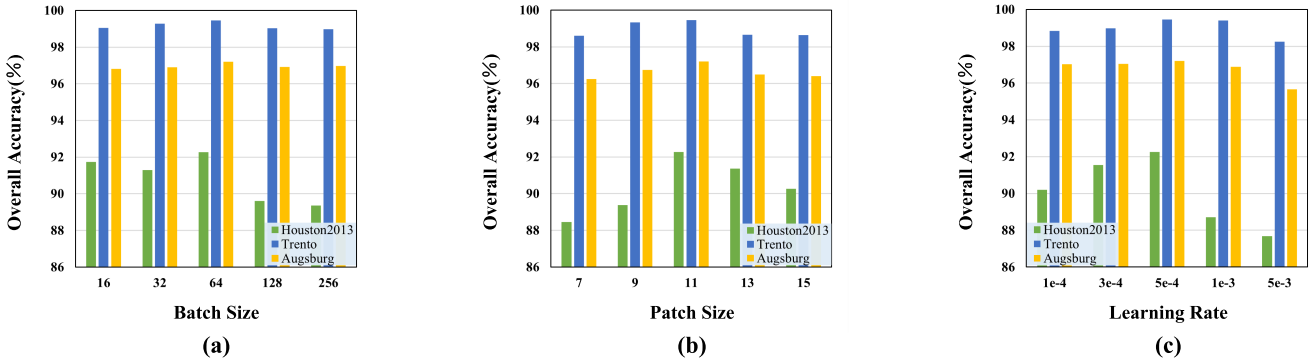


Fig. 9. Impact of different parameters on multiple datasets. (a) Batch size, (b) patch size, and (c) learning rate.

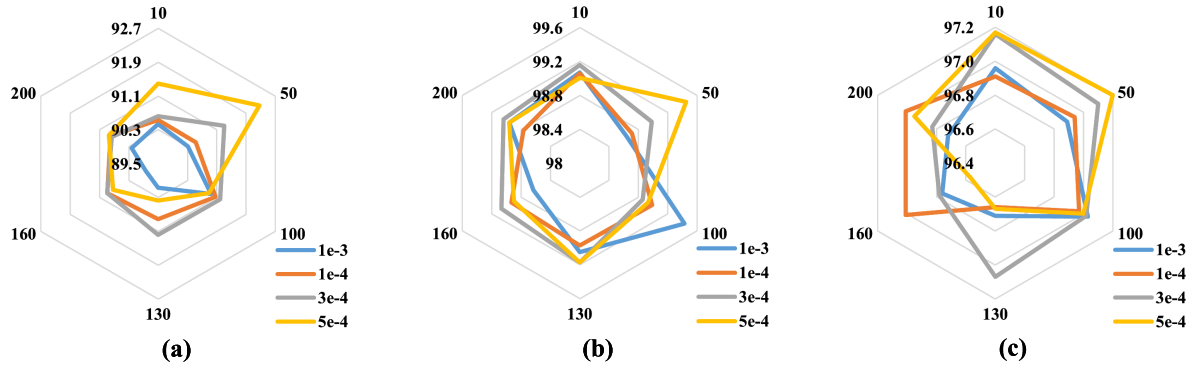


Fig. 10. Impact of different values of step size and learning rate on OA. (a) Houston2013 dataset. (b) Trento dataset. (c) Augsburg dataset. The figure contains three radar charts, where each axis represents a different step size, and the curves of different colors represent different learning rates. The farther away from the center, the larger the OA value.

D. Classification Results and Analysis

To verify the superiority of the proposed M2FNet, we compared it with nine state-of-the-art methods, including 1-D-CNN [34], 3-D-CNN [36], M3D-CNN [37], 3-D deep learning approach (3-D-DLA) [38], spectral–spatial morphological attention transformer (morphFormer) [64], a LSAF model [53], CoupledCNN [54], a HCT [55], nearest-neighbor-based contrastive learning (NNC) [56], and coupled adversarial learning (CALC) [57], where 1-D-CNN, 3-D-CNN, M3D-CNN, 3-D-DLA and morphFormer were designed for HSI classification tasks. For a fair comparison, all the compared methods adopt the default parameters provided in their works.

1) Quantitative Results and Analysis: A quantitative assessment of classification performance is presented in Tables II–IV. The best classification values are displayed in bold. The results show that the proposed approach is superior to all other techniques in terms of OA, AA, and κ and exhibits better performance in most cases in terms of classwise accuracy.

Taking Houston2013 as an example, Table II shows the generalization ability of the Houston2013 dataset for disjoint

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TABLE II

CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE HOUSTON2013 DATASET (OPTIMAL RESULTS ARE GIVEN IN BOLD)

NO.	Only HSI Input					HSI and LiDAR Input					
	1D-CNN [34]	3D-CNN [36]	M3D-CNN [37]	3D-DLA [38]	morphFormer [64]	LSAF [53]	CoupledCNN [54]	HCT [55]	NNC [56]	CALC [57]	M2FNet (proposed)
C01	81.41±0.19	83.03±6.48	82.43±0.62	82.62±0.18	**89.74±10.20**	83.09±0.00	83.06±0.09	82.86±0.17	81.22±1.33	81.21±0.56	84.12±6.17
C02	81.51±0.67	78.50±4.13	83.02±0.78	81.32±0.37	85.15±0.00	84.75±0.49	83.80±1.94	87.59±6.73	84.61±0.51	84.34±0.66	**95.24±6.04**
C03	95.57±0.41	51.88±10.63	94.85±0.90	97.09±0.82	96.33±0.70	99.60±0.22	95.51±2.77	91.16±4.16	**99.73±0.11**	91.42±1.40	98.29±0.41
C04	89.48±0.52	79.19±7.57	92.23±0.57	91.50±0.14	94.03±4.15	93.15±0.16	93.24±1.21	95.64±3.82	**98.42±1.68**	91.49±1.41	95.68±3.38
C05	90.46±1.65	86.45±3.91	99.71±0.41	99.17±0.23	99.28±0.20	99.88±0.11	99.85±0.11	99.79±0.36	99.81±0.16	**99.97±0.04**	99.12±0.68
C06	76.22±1.39	86.71±8.59	92.30±0.69	91.84±1.61	**100±0.00**	97.90±2.42	98.83±1.48	94.54±1.87	96.73±1.75	98.60±1.39	95.80±0.00
C07	55.25±0.83	80.65±7.63	84.23±2.74	83.30±1.39	89.13±0.06	**93.61±1.65**	90.66±2.15	92.79±2.08	92.84±0.42	88.12±1.75	92.22±2.26
C08	21.30±2.19	40.96±5.89	65.59±2.81	57.01±3.74	76.78±7.05	90.43±2.22	81.19±3.32	83.74±7.03	82.01±3.10	86.78±6.55	**90.90±3.10**
C09	77.99±1.39	74.15±0.47	80.51±0.61	78.53±2.74	84.32±1.33	82.43±4.14	87.12±3.89	85.51±2.38	82.59±1.87	88.85±1.86	**92.35±3.16**
C10	27.12±0.60	40.21±5.00	42.66±1.92	42.31±0.87	63.03±3.27	63.85±1.30	63.30±4.65	58.41±2.96	62.87±3.46	67.37±1.32	**76.31±4.07**
C11	56.76±1.61	61.00±8.21	65.71±2.99	66.19±2.70	91.65±0.67	85.53±4.19	90.73±3.72	93.54±2.34	82.92±1.23	**98.71±1.63**	94.44±2.84
C12	19.66±2.89	62.34±29.03	85.94±2.13	85.71±2.62	96.44±0.67	90.08±1.69	91.43±4.93	88.45±4.07	92.37±3.10	**96.91±1.49**	94.56±1.59
C13	28.42±2.19	66.78±5.46	85.61±1.40	79.29±1.26	88.42±3.96	90.78±1.52	90.09±2.90	92.28±0.00	**92.39±0.20**	89.17±2.52	91.15±1.84
C14	98.38±0.00	87.85±4.85	97.97±2.83	99.19±0.70	96.76±2.29	98.88±1.96	98.20±1.95	99.19±1.40	96.22±1.16	**99.88±0.19**	99.35±1.05
C15	96.82±0.55	39.39±12.06	98.73±0.92	98.09±0.76	95.98±5.68	99.94±0.10	97.69±3.35	**99.78±0.29**	90.13±4.67	97.88±2.62	98.26±1.85
OA(%)	63.33±0.31	67.46±1.49	80.49±0.56	79.19±0.30	88.11±0.08	88.24±0.24	87.76±0.13	88.03±0.09	87.22±0.13	89.26±0.32	**92.26±0.30**
AA(%)	66.42±0.10	67.94±2.10	83.43±0.36	82.21±0.05	89.80±0.16	90.26±0.18	89.65±0.22	89.68±0.14	88.99±0.19	90.71±0.21	**93.18±0.38**
$\kappa\times100$	60.55±0.32	64.77±1.55	78.98±0.60	77.58±0.32	87.09±0.09	87.28±0.27	86.76±0.13	87.03±0.10	86.16±0.15	88.37±0.35	**91.61±0.32**

TABLE III

CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE TRENTO DATASET (OPTIMAL RESULTS ARE GIVEN IN BOLD)

NO.	Only HSI Input					HSI and LiDAR Input					
	1D-CNN [34]	3D-CNN [36]	M3D-CNN [37]	3D-DLA [38]	morphFormer [64]	LSAF [53]	CoupledCNN [54]	HCT [55]	NNC [56]	CALC [57]	M2FNet (proposed)
C01	55.12±4.79	93.76±1.99	97.64±0.54	91.48±2.48	**99.47±0.42**	99.39±0.57	98.77±0.42	99.20±0.42	99.09±0.36	99.82±0.13	99.20±0.46
C02	68.29±5.81	79.64±1.86	81.75±1.04	76.55±3.62	93.66±1.37	96.50±0.93	93.47±3.38	97.23±1.47	97.31±0.99	**98.65±0.85**	98.15±1.04
C03	40.29±26.71	31.43±1.80	86.28±5.13	61.01±11.82	94.40±2.83	48.52±3.90	89.94±2.20	93.54±3.07	96.96±6.04	86.14±7.45	**97.11±1.70**
C04	95.76±0.35	99.98±0.03	99.45±0.63	97.38±0.52	99.98±0.01	**100±0.00**	99.99±0.01	**100±0.00**	99.97±0.03	**100±0.00**	99.99±0.01
C05	82.84±1.44	99.88±0.19	99.76±0.10	99.51±0.19	99.95±0.06	99.66±0.01	99.98±0.01	99.92±0.16	99.99±0.03	99.99±0.003	**99.99±0.004**
C06	73.87±2.92	86.63±1.27	85.18±0.38	85.52±4.87	92.01±1.48	93.61±2.53	95.03±0.81	95.42±1.77	97.56±0.74	97.33±2.35	**97.95±1.40**
OA(%)	80.03±0.45	94.67±0.06	95.91±0.16	93.51±0.50	98.37±0.01	97.97±0.27	98.51±0.24	99.01±0.12	99.30±0.19	99.34±0.23	**99.45±0.09**
AA(%)	69.36±4.36	81.89±0.20	91.68±0.63	85.24±2.03	96.58±0.39	89.61±0.72	96.19±0.52	97.55±0.39	98.48±1.06	96.99±1.24	**98.73±0.27**
$\kappa\times100$	73.21±0.62	92.87±0.08	94.54±0.21	91.32±0.67	97.83±0.02	97.30±0.36	98.02±0.33	98.69±0.16	99.07±0.25	99.12±0.31	**99.26±0.12**

TABLE IV

CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE AUGSBURG DATASET (OPTIMAL RESULTS ARE GIVEN IN BOLD)

NO.	Only HSI Input					HSI and LiDAR Input					
	1D-CNN [34]	3D-CNN [36]	M3D-CNN [37]	3D-DLA [38]	morphFormer [64]	LSAF [53]	CoupledCNN [54]	HCT [55]	NNC [56]	CALC [57]	M2FNet (proposed)
C01	73.41±3.12	96.23±2.04	98.43±0.13	95.12±0.32	98.65±0.46	**99.51±0.13**	97.77±0.72	98.75±0.29	99.45±0.17	98.37±0.55	98.80±0.36
C02	83.23±0.43	96.77±1.69	97.70±0.22	92.91±0.27	**98.98±0.22**	96.71±3.63	97.66±0.96	98.51±0.43	98.62±0.39	98.84±0.22	98.89±0.24
C03	13.81±0.41	76.65±8.20	72.44±2.76	39.21±2.15	86.90±0.85	83.43±3.90	81.31±7.20	87.11±2.73	87.97±6.80	**93.11±2.24**	90.87±2.15
C04	83.48±0.91	95.96±0.60	97.87±0.23	95.24±0.35	98.59±0.51	99.21±0.13	98.95±0.20	98.64±0.23	**99.25±0.19**	99.09±0.00	98.74±0.33
C05	0.42±0.10	24.07±23.18	33.63±8.69	18.34±2.65	83.91±2.06	55.94±3.47	48.35±12.02	64.45±5.72	**93.96±1.29**	58.50±10.08	81.83±5.60
C06	2.92±0.53	41.48±1.24	27.80±3.33	15.12±1.51	69.48±0.27	61.14±2.34	49.31±9.72	72.35±4.65	49.00±10.94	62.63±7.78	**73.13±4.23**
C07	34.54±0.62	32.80±7.84	51.32±3.17	45.07±1.49	65.54±1.63	60.82±5.82	55.12±2.93	61.40±3.27	**71.49±0.72**	60.90±1.70	70.22±2.74
OA(%)	74.96±0.43	92.46±0.36	93.80±0.07	88.33±0.10	96.81±0.07	95.65±1.37	95.11±0.55	96.51±0.14	96.85±0.01	96.76±0.17	**97.20±0.07**
AA(%)	41.69±0.43	66.28±5.34	68.46±1.92	57.29±0.65	86.01±0.58	79.54±0.75	75.50±1.23	83.03±0.72	85.68±0.56	81.63±0.20	**87.50±0.72**
$\kappa\times100$	63.02±0.71	89.13±0.615	91.03±0.11	83.11±0.15	95.43±0.10	93.77±1.92	92.95±0.79	94.99±0.20	95.48±0.01	95.35±0.25	**95.99±0.10**

training and test samples. First, methods that rely solely on HSI tend to exhibit lower performance compared with those that integrate LiDAR data, highlighting the beneficial effect of incorporating elevation information. Among the array of HSI-only approaches, while morphFormer records the highest classification accuracy, M2FNet surpasses morphFormer by significant margins of 4.15% in OA, 3.38% in AA, and 4.52% in κ. Moreover, our proposed method maintains a competitive edge over other fusion-based classification techniques that use both HSI and LiDAR data, underscoring its efficacy in leveraging multisource information for enhanced classification outcomes. Specifically, M2FNet outperforms CALC by 3.00% in terms of OA, 2.47% in terms of AA, and 3.24% in terms of κ. It is not difficult to see that the classification accuracy of

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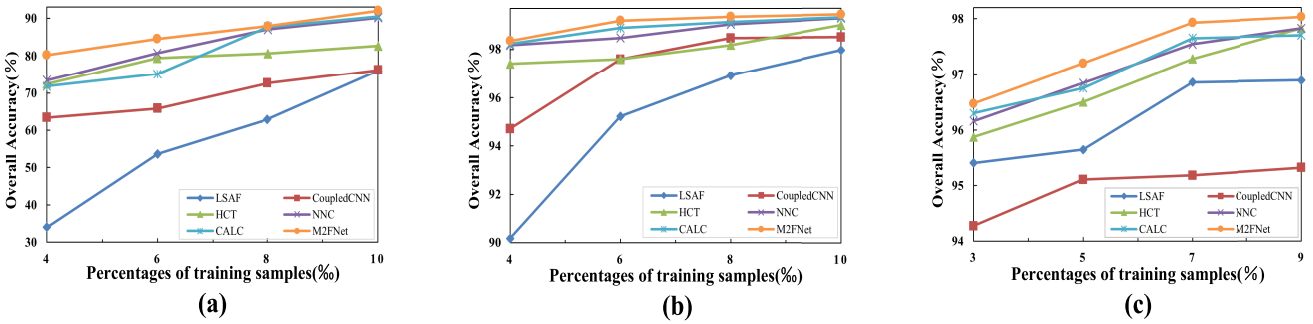


Fig. 11. OA of different models with different training samples percentages. (a) Houston2013 dataset. (b) Trento dataset. (c) Augsburg dataset.

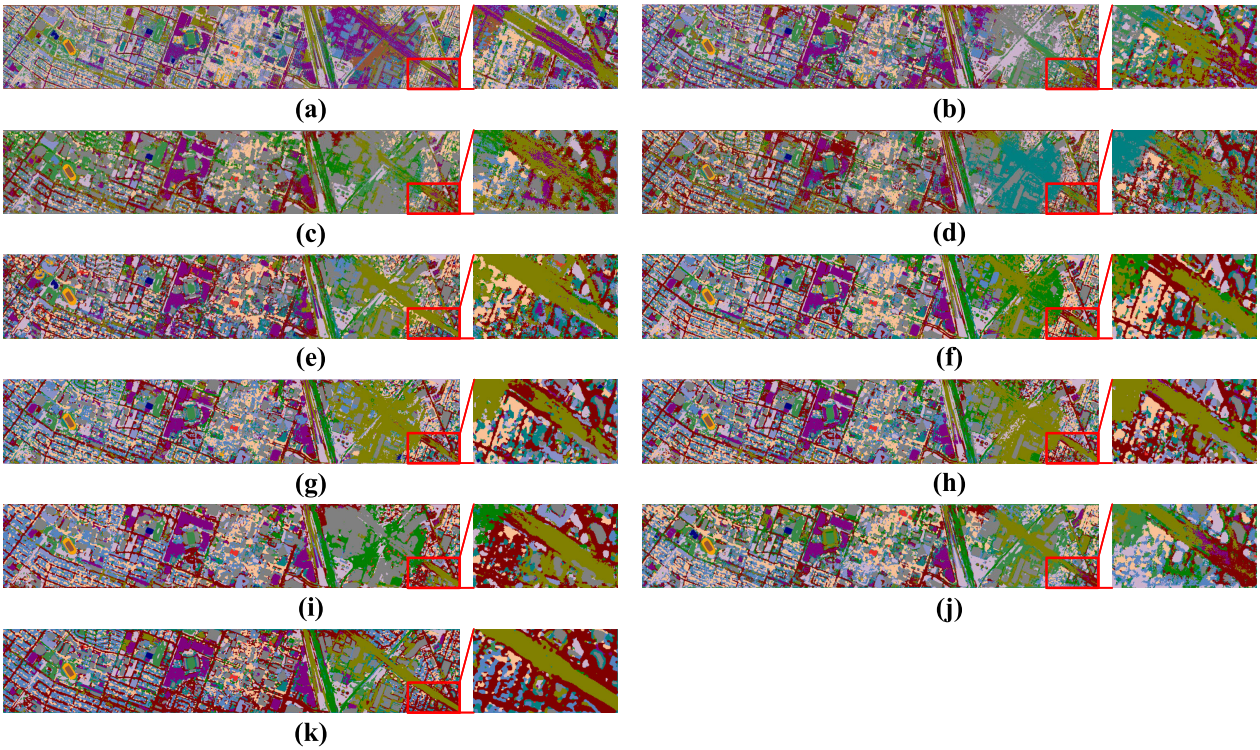


Fig. 12. Classification maps for the Houston2013 dataset. (a) 1-D-CNN. (b) 3-D-CNN. (c) M3D-CNN. (d) 3-D-DLA. (e) morphFormer. (f) LSAF. (g) CoupledCNN. (h) HCT. (i) NNC. (j) CALC. (k) Proposed M2FNet.

M2FNet for "Health grass," "Stressed grass," "Commercial," "Road," and "Highway" is the highest among the compared methods. This is closely related to the multiscale feature extraction, FE, and deep fusion of features in our method. Similar conclusions can also be drawn from Tables III and IV. These two tables present the quantitative results of the comparative methods on the Trento and Augsburg datasets. It is evident that the proposed M2FNet significantly outperforms the comparative methods in terms of OA, AA, and k metrics. This once again confirms the superiority of the proposed M2FNet.

To measure the stability and robustness of the proposed method with different percentages of training samples, we compared it with LSAF, CoupledCNN, HCT, NNC, and CALC using different training sample sizes. About 0.4%, 0.6%, 0.8%, and 1% labeled samples were randomly selected as training data for the Houston2013 and Trento datasets. And 3%, 5%, 7%, and 9% labeled samples were randomly selected as training data for the Augsburg datasets. The experimental results are shown in Fig. 11. It is clear that OA increases as the number of samples increases. The proposed method still maintains good performance under the condition of ultrasmall sample size, which reflects the superior feature information capture ability of M2FNet. This is inseparable from the unique backbone module and FE module. In addition, we observed that some of the methods compared performed poorly with ultrasmall sample sizes.

2) Visual Evaluation and Analysis: Figs. 12–14 illustrate visual maps for qualitative evaluation of the classification results. The results show that the proposed M2FNet provides a more accurate noise-free classification map. This can visualize the experimental results mentioned above. We can significantly observe the differences among various methods.

The classification maps of single data sources and simple deep convolutional classification networks have more misclassification phenomena, accompanied by serious noise. For LSAF, CoupledCNN, HCT, NNC, and CALC, misclassification and noise phenomena are reduced, but the classification accuracy is still low for some categories. For example, for the Houston2013 dataset, it is easy to see from the classification

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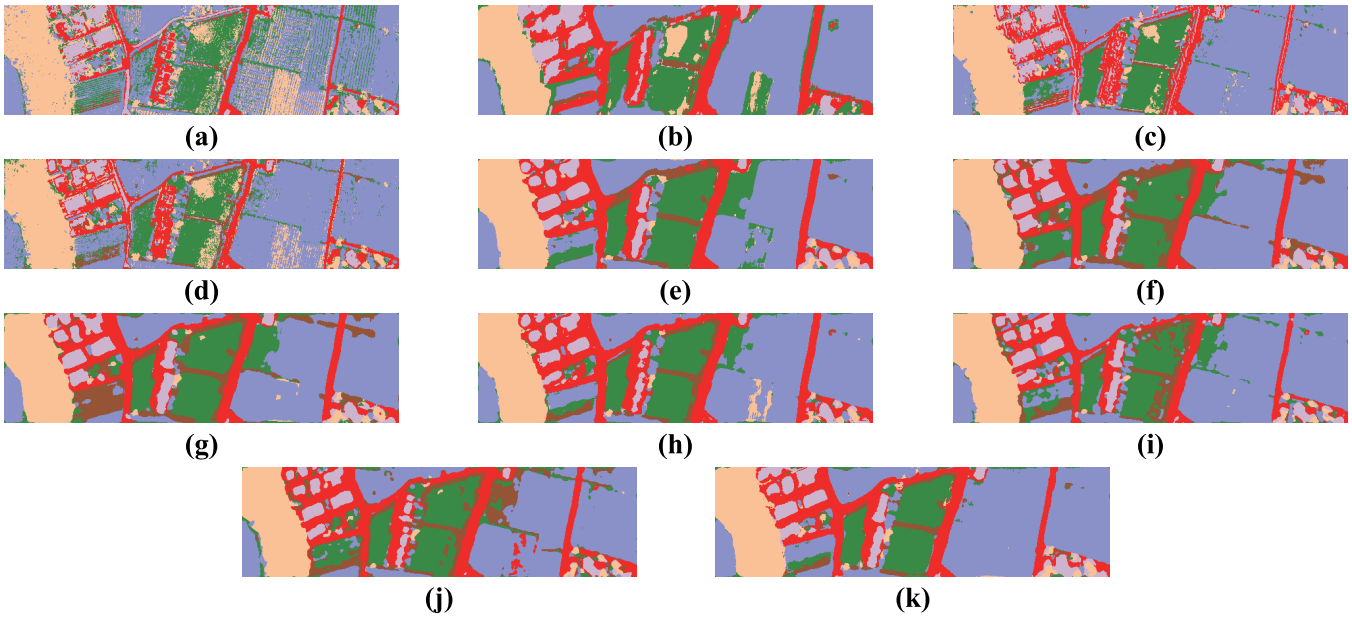


Fig. 13. Classification maps for the Trento dataset. (a) 1-D-CNN. (b) 3-D-CNN. (c) M3D-CNN. (d) 3-D-DLA. (e) morphFormer. (f) LSAF. (g) CoupledCNN. (h) HCT. (i) NNC. (j) CALC. (k) Proposed M2FNet.

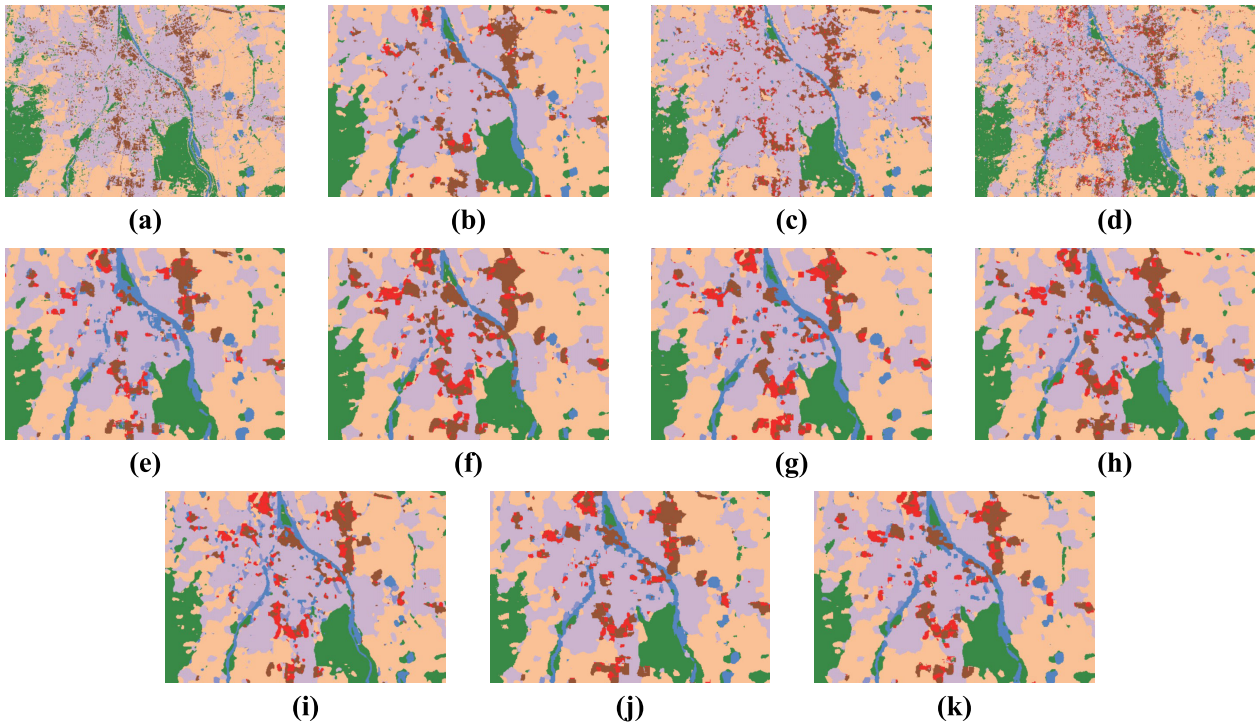


Fig. 14. Classification maps for the Augsburg dataset. (a) 1-D-CNN. (b) 3-D-CNN. (c) M3D-CNN. (d) 3-D-DLA. (e) morphFormer. (f) LSAF. (g) CoupledCNN. (h) HCT. (i) NNC. (j) CALC. (k) Proposed M2FNet.

TABLE V

ABLATION ANALYSIS OF DIFFERENT MODAL DATA INPUTS (OPTIMAL RESULTS ARE GIVEN IN BOLD)

Cases	Houston2013			Trento			Augsburg		
	OA (%)	AA (%)	$\kappa \times 100$	OA (%)	AA (%)	$\kappa \times 100$	OA (%)	AA (%)	$\kappa \times 100$
Only HSI	88.32	90.2	87.34	98.02	96.26	97.36	96.6	84.04	95.13
Only LiDAR	55.22	55.19	51.51	97.63	91.79	96.84	90.19	69.59	85.89
HSI+LiDAR	**92.26**	**93.18**	**91.61**	**99.45**	**98.73**	**99.26**	**97.20**	**87.50**	**95.99**

map that various methods misclassify the highway category seriously, and the classification map of M2FNet shows the results closest to the ground-truth map. In the Trento dataset, the data have more homogeneous and larger objects than the Houston dataset and are easier to classify. Each classifier obtains high classification accuracy, but it is not difficult to find the high accuracy of M2FNet for the classification of Ground and Roads categories through the classification map. For

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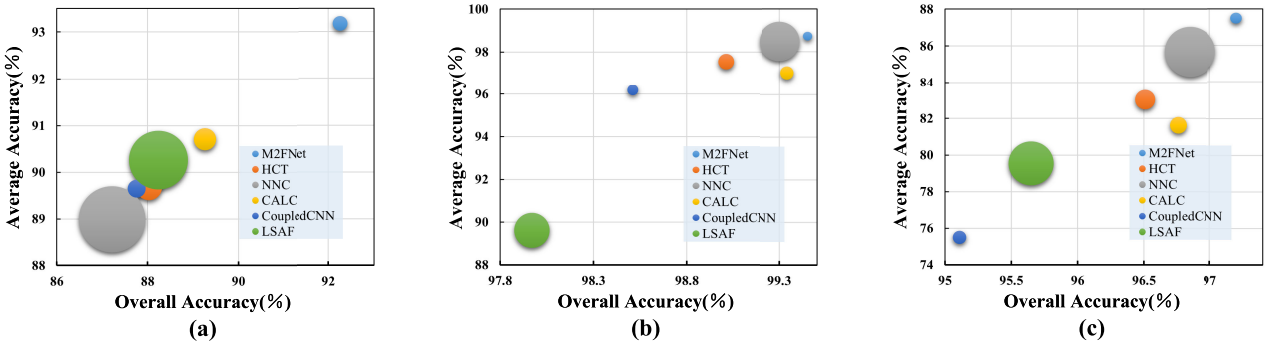


Fig. 15. Comparison of different models in terms of OA, AA, and training parameter count. (a) Houston2013 dataset. (b) Trento dataset. (c) Augsburg dataset. This chart consists of three bubble plots, with each bubble plot representing OA values on the x-axis and AA values on the y-axis. It is important to note that the center coordinates of the bubbles represent the OA and AA values of the models, while the size of the bubbles indicates the training parameter count of different models.

TABLE VI
ABLATION ANALYSIS OF DIFFERENT COMPONENTS IN THE PROPOSED MODEL ON THE HOUSTON2013 DATASET (OPTIMAL RESULTS ARE GIVEN IN BOLD)

Cases	Component					Indicators		
	Conv3-D	MConv2-D	FE	DConv	FTE	OA (%)	AA (%)	$\kappa \times 100$
01	✓	✓	✓	MHSA	✓	91.83	92.86	91.14
02	✓	✓	✗	✓	✓	90.61	91.63	89.82
03	✓	Conv2-D	✓	✓	✓	90.93	91.71	90.17
04	✓	✗	✗	✗	✗	82.20	83.06	80.73
05	✓	✓	✗	✗	✗	87.94	88.86	86.94
06	✓	✓	✓	✗	✗	89.62	90.81	88.76
07	✓	✓	✓	✓	✗	90.96	91.98	90.20
08	✓	✓	✓	✓	✓	**92.26**	**93.18**	**91.61**

the Augsburg dataset, we note that the classification method without incorporating elevation information does not perform well in terms of classification accuracy for the Allotment and Commercial Area categories. In the image of the classification map, M2FNet shows high accuracy in the classification of Industrial Area and Commercial Area categories.

In conclusion, the comparison of several state-of-the-art classifiers in visual effect still validates that the proposed M2FNet has excellent classification performance.

3) Computational Complexity Analysis: In Fig. 15(a)–(c), the training parameters of the proposed method are compared with those of a deep network that fuses multisource data for hyperspectral classification. Specifically, we show the number of OA, AA, and training parameters for the Houston2013, Trento, and Augsburg datasets. The coordinates of the center of the circle in the figure represent OA and AA, and the number of parameters is represented by the radius of the circle. The efficiency of M2FNet is evident in the Houston2013 and Augsburg datasets, where it requires fewer parameters and exhibits the highest accuracy. For the Trento dataset, although the high accuracy of M2FNet performs insignificantly and is close to the accuracy of NNCNet, the number of training parameters of NNCNet is huge than our proposed method.

E. Ablation Analysis

1) Ablation Analysis of Different Modal Data Inputs: We conducted two sets of experiments, one using only HSI data as input and the other using only LiDAR data as input. According to the experimental results in Table V, our fusion network achieves better classification results on the three datasets, manifested by higher OA and AA values. This shows that our proposed fusion network can make full use of the effective information of different modal data, and these data play a positive role in improving classification performance.

2) Ablation Analysis of Different Components: To fully demonstrate the specific impact of each component of the proposed M2FNet on the classification performance, we conduct ablation experiments on the Houston2013 dataset. We experimented with the following five different components: Conv3-D and multiscale Conv2-D (MConv2-D) used in the feature extraction module, FE module in DConvformer, deep dilated convolution anti-bottleneck module (DConv), and feature token exchange (FTE) module.

As listed in Table VI, the Houston 2013 dataset is used to evaluate seven variants of the proposed framework with different combinations of components removed. The classification accuracy of these variants was analyzed to assess the impact of various components on the model. The first variant is special compared with the third variant. In the first variant, we use the multihead self-attention (MHSA) module instead of the DConv component, and in the third variant, we replace MConv2-D with normal Conv2-D. Moreover, we note from the results of variants 04–07 that the OA is 82.20% when we keep only the Conv3-D component. MConv2-D, FE, DConv, and FTE components improve OA by 5.74%, 1.68%, 1.34%, and 1.30%, respectively. This analysis further confirms the effectiveness of the proposed network framework, showing that all its main components are important.

IV. CONCLUSION

In this article, we propose a novel multiscale CNN and transformer network for accurate joint classification of HSI and LiDAR data. A two-branch network framework is designed by combining CNN-based feature extraction and transformer-based feature learning to achieve efficient feature mining. Specifically, a multiscale 3-D–2-D mixed CNN feature extraction method is used to create a novel DConvformer encoder to fuse the two types of features in the fusion stage. The experimental results on three groups of HSI and LiDAR data show that the proposed method has a superior classification effect compared with the existing state-of-the-art classifiers. In summary, the proposed M2FNet can extract more discriminative information and effectively fuse heterogeneous data features for HSI and LiDAR joint classification.

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In the future, we can use more innovative multiscale feature extraction architectures (e.g., cross-transformer) and use lightweight operation (e.g., alpha-beta pruning) to enhance cross-modal information fusion. It will improve the network's ability to integrate multimodal data, further enhancing performance and reducing complexity.

REFERENCES

[1] Y. Duan, F. Luo, M. Fu, Y. Niu, and X. Gong, "Classification via structure-preserved hypergraph convolution network for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507113.

[2] J. Yue, L. Fang, and M. He, "Spectral–spatial latent reconstruction for open-set hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 5227–5241, 2022.

[3] L. Sun et al., "Low rank component induced spatial–spectral kernel method for hyperspectral image classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 10, pp. 3829–3842, Oct. 2020.

[4] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[5] S. Sudharsan, R. Hemalatha, and S. Radha, "A survey on hyperspectral imaging for mineral exploration using machine learning algorithms," in *Proc. Int. Conf. Wireless Commun. Signal Process. Netw. (WiSPNET)*, Mar. 2019, pp. 206–212.

[6] P. Gong, Z. Li, H. Huang, G. Sun, and L. Wang, "ICESat GLAS data for urban environment monitoring," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 3, pp. 1158–1172, Mar. 2011.

[7] X. Huang, H. Liu, and L. Zhang, "Spatiotemporal detection and analysis of urban villages in mega city regions of China using high-resolution remotely sensed imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3639–3657, Jul. 2015.

[8] P. Kempeneers, F. Sedano, L. Seebach, P. Strobl, and J. San-Miguel-Ayanz, "Data fusion of different spatial resolution remote sensing images applied to forest-type mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 12, pp. 4977–4986, Dec. 2011.

[9] D. Murugan, A. Garg, and D. Singh, "Development of an adaptive approach for precision agriculture monitoring with drone and satellite data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 12, pp. 5322–5328, Dec. 2017.

[10] L. Karthikeyan, I. Chawla, and A. K. Mishra, "A review of remote sensing applications in agriculture for food security: Crop growth and yield, irrigation, and crop losses," *J. Hydrol.*, vol. 586, Jul. 2020, Art. no. 124905.

[11] X. Wei, W. Li, M. Zhang, and Q. Li, "Medical hyperspectral image classification based on end-to-end fusion deep neural network," *IEEE Trans. Instrum. Meas.*, vol. 68, no. 11, pp. 4481–4492, Nov. 2019.

[12] H. Gao, Y. Zhang, Z. Chen, S. Xu, D. Hong, and B. Zhang, "A multi-depth and multibranch network for hyperspectral target detection based on band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5506818.

[13] H. Gao, Y. Zhang, Z. Chen, F. Xu, D. Hong, and B. Zhang, "Hyperspectral target detection via spectral aggregation and separation network with target band random mask," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515516.

[14] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "Deep learning classifiers for hyperspectral imaging: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, pp. 279–317, Dec. 2019.

[15] J. Bai, B. Ding, Z. Xiao, L. Jiao, H. Chen, and A. C. Regan, "Hyperspectral image classification based on deep attention graph convolutional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5504316.

[16] Z. Gao, B. Pan, X. Xu, T. Li, and Z. Shi, "LiCa: Label-indicate-conditional-alignment domain generalization for pixel-wise hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5519011.

[17] Z. Wang and M. Menenti, "Challenges and opportunities in LiDAR remote sensing," *Frontiers Remote Sens.*, vol. 2, Mar. 2021, Art. no. 641723.

[18] M. Zhang, W. Li, Q. Du, L. Gao, and B. Zhang, "Feature extraction for classification of hyperspectral and LiDAR data using patch-to-patch CNN," *IEEE Trans. Cybern.*, vol. 50, no. 1, pp. 100–111, Jan. 2020.

[19] M. Zhang, W. Li, Y. Zhang, R. Tao, and Q. Du, "Hyperspectral and LiDAR data classification based on structural optimization transmission," *IEEE Trans. Cybern.*, vol. 53, no. 5, pp. 3153–3164, May 2023.

[20] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[21] Q. Ye, P. Huang, Z. Zhang, Y. Zheng, L. Fu, and W. Yang, "Multiview learning with robust double-sided twin SVM," *IEEE Trans. Cybern.*, vol. 52, no. 12, pp. 12745–12758, Dec. 2022.

[22] J. Ham, Y. Chen, M. M. Crawford, and J. Ghosh, "Investigation of the random forest framework for classification of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 492–501, Mar. 2005.

[23] Y. Guo, S. Han, Y. Li, C. Zhang, and Y. Bai, "K-nearest neighbor combined with guided filter for hyperspectral image classification," *Proc. Comput. Sci.*, vol. 129, pp. 159–165, Jan. 2018.

[24] C. Rodarmel and J. Shan, "Principal component analysis for hyperspectral image classification," *Surv. Land Inf. Syst.*, vol. 62, no. 2, pp. 115–122, Jun. 2002.

[25] T. V. Bandos, L. Bruzzone, and G. Camps-Valls, "Classification of hyperspectral images with regularized linear discriminant analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 3, pp. 862–873, Mar. 2009.

[26] W. Li, C. Chen, H. Su, and Q. Du, "Local binary patterns and extreme learning machine for hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3681–3693, Jul. 2015.

[27] M. Colgan, C. Baldeck, J.-B. Féret, and G. Asner, "Mapping savanna tree species at ecosystem scales using support vector machine classification and BRDF correction on airborne hyperspectral and LiDAR data," *Remote Sens.*, vol. 4, no. 11, pp. 3462–3480, Nov. 2012.

[28] R. Huang and J. Zhu, "Using random forest to integrate LiDAR data and hyperspectral imagery for land cover classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2013, pp. 3978–3981.

[29] S. Jia et al., "Multiple feature-based superpixel-level decision fusion for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1437–1452, Feb. 2021.

[30] S. Indolia, A. K. Goswami, S. P. Mishra, and P. Asopa, "Conceptual understanding of convolutional neural network—A deep learning approach," *Proc. Comput. Sci.*, vol. 132, pp. 679–688, Jan. 2018.

[31] H. Gao, Y. Yang, C. Li, L. Gao, and B. Zhang, "Multiscale residual network with mixed depthwise convolution for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3396–3408, Apr. 2021.

[32] H. Gao, Z. Chen, and F. Xu, "Adaptive spectral–spatial feature fusion network for hyperspectral image classification using limited training samples," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 107, Mar. 2022, Art. no. 102687.

[33] H. Yu et al., "Global spatial and local spectral similarity-based manifold learning group sparse representation for hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3043–3056, May 2020.

[34] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, pp. 1–12, Jan. 2015.

[35] W. Zhao and S. Du, "Spectral–spatial feature extraction for hyperspectral image classification: A dimension reduction and deep learning approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4544–4554, Aug. 2016.

[36] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[37] M. He, B. Li, and H. Chen, "Multi-scale 3D deep convolutional neural network for hyperspectral image classification," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2017, pp. 3904–3908.

[38] A. B. Hamida, A. Benoit, P. Lambert, and C. B. Amar, "3-D deep learning approach for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4420–4434, Aug. 2018.

[39] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30. Red Hook, NY, USA: Curran Associates, 2017, pp. 5998–6008.

[40] L. Fu, D. Zhang, and Q. Ye, "Recurrent thrifty attention network for remote sensing scene recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8257–8268, Oct. 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:28:07 UTC from IEEE Xplore. Restrictions apply.

[41] T. Guo, R. Wang, F. Luo, X. Gong, L. Zhang, and X. Gao, "Dual-view spectral and global spatial feature fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5512913.
[42] H. Zhou, F. Luo, H. Zhuang, Z. Weng, X. Gong, and Z. Lin, "Attention multihop graph and multiscale convolutional fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508614.
[43] N. Chen, J. Yue, L. Fang, and S. Xia, "SpectralDiff: A generative framework for hyperspectral image classification with diffusion models," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522416.
[44] A. Dosovitskiy et al., "An image is worth 16 × 16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
[45] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.
[46] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.
[47] Y. Fang, Q. Ye, L. Sun, Y. Zheng, and Z. Wu, "Multiattention joint convolution feature representation with lightweight transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5513814.
[48] B. Tu, X. Liao, Q. Li, Y. Peng, and A. Plaza, "Local semantic feature aggregation-based transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536115.
[49] Y. Su et al., "ACGT-Net: Adaptive cuckoo refinement-based graph transfer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5521314.
[50] Y. Su, L. Gao, M. Jiang, A. Plaza, X. Sun, and B. Zhang, "NSCKL: Normalized spectral clustering with kernel-based learning for semisupervised hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 53, no. 10, pp. 6649–6662, Oct. 2023.
[51] B. Tu, W. He, Q. Li, Y. Peng, and A. Plaza, "A new context-aware framework for defending against adversarial attacks in hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5505114.
[52] M. Zhang, X. Zhao, W. Li, Y. Zhang, R. Tao, and Q. Du, "Cross scene joint classification of multisource data with multilevel domain adaptation network," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, pp. 1–13, Apr. 6, 2023, doi: 10.1109/TNNLS.2023.3262599.
[53] M. Feng, F. Gao, J. Fang, and J. Dong, "Hyperspectral and LiDAR data classification based on linear self-attention," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jun. 2021, pp. 2401–2404.
[54] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.
[55] G. Zhao, Q. Ye, L. Sun, Z. Wu, C. Pan, and B. Jeon, "Joint classification of hyperspectral and LiDAR data using a hierarchical CNN and transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500716.
[56] M. Wang, F. Gao, J. Dong, H.-C. Li, and Q. Du, "Nearest neighbor-based contrastive learning for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501816.
[57] T. Lu, K. Ding, W. Fu, S. Li, and A. Guo, "Coupled adversarial learning for fusion classification of hyperspectral and LiDAR data," *Inf. Fusion*, vol. 93, pp. 118–131, May 2023.
[58] B. Du, Z. Liu, and F. Luo, "Deep multi-scale attention network for RNA-binding proteins prediction," *Inf. Sci.*, vol. 582, pp. 287–301, Jan. 2022.
[59] R. Dai, S. Das, K. Kahatapitiya, M. S. Ryoo, and F. Brémond, "MS-TCT: Multi-scale temporal ConvTransformer for action detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 20041–20051.
[60] H. Idrees, I. Saleemi, C. Seibert, and M. Shah, "Multi-source multi-scale counting in extremely dense crowd images," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2013, pp. 2547–2554.
[61] T.-Y. Ma, H.-C. Li, R. Wang, Q. Du, X. Jia, and A. Plaza, "Lightweight tensorized neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5544816.
[62] W.-J. Guo, W. Xie, K. Jiang, Y. Li, J. Lei, and L. Fang, "Toward stable, interpretable, and lightweight hyperspectral super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 22272–22281.
[63] S. K. Roy, R. Mondal, M. E. Paoletti, J. M. Haut, and A. Plaza, "Morphological convolutional neural networks for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 8689–8702, 2021.
[64] S. K. Roy, A. Deria, C. Shah, J. M. Haut, Q. Du, and A. Plaza, "Spectral–spatial morphological attention transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5503615.



Le Sun (Member, IEEE) received the B.S. degree in information and computing science and the Ph.D. degree in computer science from Nanjing University of Science and Technology (NJUST), Nanjing, Jiangsu, China, in 2009 and 2014, respectively.

From 2015 to 2018, he was a Post-Doctoral Researcher with the School of Electronic and Electrical Engineering, Sungkyunkwan University Natural Science Campus, Suwon, South Korea. He is currently a Full Professor at the School of Computer Science, NJUST. His research interests include high-dimensional signal processing and deep learning.



Xinyu Wang received the B.S. degree in computer science and technology from the Kexin College, Hebei University of Engineering, Handan, China, in 2021. He is currently pursuing the M.S. degree in software engineering with Nanjing University of Information Science and Technology, Nanjing, China.

His research interests include hyperspectral image processing.



Yuhui Zheng (Senior Member, IEEE) received the B.S. degree in chemistry and the Ph.D. degree in computer science from Nanjing University of Science and Technology (NJUST), Nanjing, Jiangsu, China, in 2004 and 2009, respectively.

He is currently a Full Professor at the School of Computer and Science, NJUST. His research interests cover image processing, pattern recognition, and remote sensing information systems.



Zebin Wu (Senior Member, IEEE) received the B.S. and Ph.D. degrees from the School of Computer Science and Engineering, Nanjing University of Science and Technology (NJUST), Nanjing, Jiangsu, China, in 2003 and 2008, respectively.

He is currently a Full Professor at the School of Computer Science and Engineering, NJUST. His research interests include remote sensing information processing and distributed computing.



Liyong Fu received the B.S. degree in forestry from Shanxi Agricultural University, Jinzhong, China, in 2007, the M.S. degree in forest biometrics from Nanjing Forestry University, Nanjing, China, in 2009, and the Ph.D. degree in forest biometrics from the Chinese Academy of Forestry, Beijing, China, in 2012.

He is currently a Full Professor at the Department of Forest Management and Statistics, Chinese Academy of Forestry. He has authored or coauthored a total of 43 science citation index (SCI) articles in prestigious peer-reviewed international journals including *Briefings in Bioinformatics*, *Neural Networks*, IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP), IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), and *Frontiers in Plant Science* during the recent five years.

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